

MARTI301

AI in Investment and Competitiveness

EU Business School · Masters Level 7 · Term 3 · 3 ECTS

Framework Sheet Library

Master Index · 24 Slibraries · 120 Reference Sheets

6

Course Units

covering strategy, ML, CX, finance, IP, investment

24

Slibraries (thematic clusters)

4 per unit, colour-coded

120

Framework Reference Sheets

5 per Slibrary, print-ready A4

480+

Harvard-referenced sources

peer-reviewed and practitioner

Each sheet follows a focused five-section layout: a professional definition, a one-line tagline, a custom illustrative figure, four key details, and Harvard-referenced sources. The structure enables students and practitioners to navigate the library as a reference work — treating the 120 sheets as a mental Slibrary (per Meier, 2025) that scales from single-framework lookup to full strategic reasoning across multiple connected tools.

Dr. Hildegard Haas

EU Business School · MARTI301

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Unit 1

AI Strategy, Governance & Competitive Advantage

Foundations · Klarna anchor case · 4 Slibraries · 20 frameworks

SLIBRARY 01 · AI STRATEGY & GOVERNANCE

5 sheets

- 1 EU AI Act Risk Pyramid *p. 9*
- 2 NIST AI RMF Lifecycle *p. 10*
- 3 Perez Techno-Economic S-Curve *p. 11*
- 4 AI Maturity Staircase *p. 12*
- 5 McKinsey AI High Performers *p. 13*

SLIBRARY 02 · COMPETITIVE ADVANTAGE

5 sheets

- 1 Porter's Five Forces *p. 14*
- 2 VRIO Framework *p. 15*
- 3 Porter's Value Chain *p. 16*
- 4 Resource-Based View *p. 17*
- 5 Disruption S-Curves *p. 18*

SLIBRARY 03 · AI PROJECT LIFECYCLE

5 sheets

- 1 CRISP-DM Cycle *p. 19*
- 2 DDED Stage-Gate *p. 20*
- 3 Build vs. Buy 2×2 *p. 21*
- 4 AI Failure Funnel *p. 22*
- 5 Data Readiness Ladder *p. 23*

SLIBRARY 04 · HUMAN-AI COLLABORATION

5 sheets

- 1 Prediction-Judgment Split *p. 24*
- 2 Centaur Model *p. 25*
- 3 Automation-Augmentation Spectrum *p. 26*
- 4 BCG People Investment *p. 27*
- 5 Human-in-the-Loop Feedback *p. 28*

Unit 2

ML & Data-Driven Decisions

Live data lab · 4 Slibraries · 20 frameworks

SLIBRARY 05 · ML MODELS & TECHNIQUES

5 sheets

- | | | |
|---|----------------------------|-------|
| 1 | ML Learning Types Taxonomy | p. 29 |
| 2 | Bias-Variance Tradeoff | p. 30 |
| 3 | Confusion Matrix 2×2 | p. 31 |
| 4 | Feature Importance Chart | p. 32 |
| 5 | Model Selection Flowchart | p. 33 |

SLIBRARY 06 · DECISION INTELLIGENCE

5 sheets

- | | | |
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| 1 | OODA Loop | p. 34 |
| 2 | Weighted Decision Matrix | p. 35 |
| 3 | Cynefin Framework | p. 36 |
| 4 | IDSS Architecture Layers | p. 37 |
| 5 | Prediction Machines Model | p. 38 |

SLIBRARY 07 · DATA STRATEGY

5 sheets

- | | | |
|---|-------------------------|-------|
| 1 | Data Value Chain | p. 39 |
| 2 | DIKW Pyramid | p. 40 |
| 3 | Data Flywheel | p. 41 |
| 4 | Data Quality Dimensions | p. 42 |
| 5 | ETL Pipeline Diagram | p. 43 |

SLIBRARY 08 · AI-DRIVEN MARKETING

5 sheets

- | | | |
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| 1 | AI Marketing Funnel | p. 44 |
| 2 | RFM Segmentation Matrix | p. 45 |
| 3 | A/B Testing Framework | p. 46 |
| 4 | Recommendation Engine | p. 47 |
| 5 | Customer Lifetime Value | p. 48 |

Unit 3

Enhanced Customer Experience

Role play · 4 Slibraries · 20 frameworks

SLIBRARY 09 · CUSTOMER JOURNEY & CX

5 sheets

- 1 Omnichannel Journey Map p. 49
- 2 CX Experience Pyramid p. 50
- 3 Service Blueprint p. 51
- 4 Touchpoint Matrix p. 52
- 5 Moments of Truth Timeline p. 53

SLIBRARY 10 · CONVERSATIONAL AI & NLP

5 sheets

- 1 NLP Pipeline Diagram p. 54
- 2 Intent-Entity-Context Model p. 55
- 3 Chatbot Decision Tree p. 56
- 4 Sentiment Analysis Spectrum p. 57
- 5 Human Escalation Flowchart p. 58

SLIBRARY 11 · CX METRICS & BUSINESS IMPACT

5 sheets

- 1 NPS-CSAT-CES Triangle p. 59
- 2 Churn Waterfall Chart p. 60
- 3 CLV-to-CAC Ratio p. 61
- 4 Loyalty Loop Diagram p. 62
- 5 CX-to-Revenue Bridge p. 63

SLIBRARY 12 · MARKETING 6.0 & IMMERSIVE

5 sheets

- 1 Marketing Evolution Timeline p. 64
- 2 Phygital Experience Model p. 65
- 3 Metaverse CX Framework p. 66
- 4 Spatial Computing Stack p. 67
- 5 Kotler 6.0 Ecosystem Map p. 68

Unit 4

The Financial Side of AI

Mini-hackathon · 4 Slibraries · 20 frameworks

SLIBRARY 13 · AI IN FINANCIAL MARKETS

5 sheets

- 1 FinTech Ecosystem Map p. 69
- 2 AI Trading Architecture p. 70
- 3 Democratisation Ladder p. 71
- 4 Startup Funding Funnel p. 72
- 5 RegTech Compliance Loop p. 73

SLIBRARY 14 · AI RISK MANAGEMENT

5 sheets

- 1 Three Lines of Defence p. 74
- 2 Model Risk Taxonomy p. 75
- 3 Stress Testing Framework p. 76
- 4 Algorithmic Bias Audit p. 77
- 5 Black Swan Radar p. 78

SLIBRARY 15 · TALENT & RESOURCES

5 sheets

- 1 AI Talent Gap Heatmap p. 79
- 2 Build-Buy-Borrow Matrix p. 80
- 3 Skills Adjacency Map p. 81
- 4 Total Cost of AI Ownership p. 82
- 5 Org Design for AI Teams p. 83

SLIBRARY 16 · SUSTAINABLE & GREEN AI

5 sheets

- 1 AI Carbon Footprint Stack p. 84
- 2 Green AI vs. Red AI p. 85
- 3 ESG-AI Integration Matrix p. 86
- 4 Compute Efficiency Curve p. 87
- 5 Responsible AI Scorecard p. 88

Unit 5

IP & Value Creation through AI

Moot court · 4 Slibraries · 20 frameworks

SLIBRARY 17 · AI & COPYRIGHT

5 sheets

- | | | |
|---|-----------------------------|-------|
| 1 | Authorship Decision Tree | p. 89 |
| 2 | AI-Generated Works Spectrum | p. 90 |
| 3 | TDM Exception Flowchart | p. 91 |
| 4 | Fair Use Four-Factor Test | p. 92 |
| 5 | Global Copyright Comparison | p. 93 |

SLIBRARY 18 · DATA AS IP

5 sheets

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| 1 | Data IP Protection Layers | p. 94 |
| 2 | Sui Generis Database Rights | p. 95 |
| 3 | Trade Secret vs. Patent 2×2 | p. 96 |
| 4 | Data Licensing Model Map | p. 97 |
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SLIBRARY 19 · IP STRATEGY & COMPETITION

5 sheets

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| 1 | Freedom-to-Operate Analysis | p. 99 |
| 2 | Patent Landscape Map | p. 100 |
| 3 | IP Portfolio Valuation | p. 101 |
| 4 | Competition-IP Tension Model | p. 102 |
| 5 | FRAND Licensing Diagram | p. 103 |

SLIBRARY 20 · LIABILITY & PROTECTION

5 sheets

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| 1 | AI Liability Chain | p. 104 |
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| 3 | Insurance & Indemnity Framework | p. 106 |
| 4 | Contractual IP Allocation | p. 107 |
| 5 | Enforcement Decision Flowchart | p. 108 |

Unit 6

Finance & Investment Plans

Business challenge · capstone · 4 Slibraries · 20 frameworks

SLIBRARY 21 · AI IN INVESTMENT MANAGEMENT

5 sheets

- | | | |
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| 1 | Investment Mgmt Value Chain | p. 109 |
| 2 | AI Use Case Taxonomy (Finance) | p. 110 |
| 3 | Robo-Advisory Architecture | p. 111 |
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| 5 | Muldowney ML Adoption Model | p. 113 |

SLIBRARY 22 · PORTFOLIO OPTIMISATION & ML

5 sheets

- | | | |
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| 1 | MPT Efficient Frontier | p. 114 |
| 2 | Black-Litterman Model | p. 115 |
| 3 | ML-Enhanced Risk-Return 2x2 | p. 116 |
| 4 | Factor Investing Architecture | p. 117 |
| 5 | Backtesting Validation Loop | p. 118 |

SLIBRARY 23 · FINANCIAL ANALYSIS & MODELLING

5 sheets

- | | | |
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| 1 | AI Financial Analysis Stack | p. 119 |
| 2 | Sentiment-Driven Alpha | p. 120 |
| 3 | Alternative Data Integration | p. 121 |
| 4 | NLP for Earnings Calls | p. 122 |
| 5 | AI-Driven Risk Model Layers | p. 123 |

SLIBRARY 24 · PLAN-TO-IMPLEMENTATION

5 sheets

- | | | |
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| 1 | Planning vs. Implementation Gap | p. 124 |
| 2 | AI Readiness Assessment Radar | p. 125 |
| 3 | Change Management Flywheel | p. 126 |
| 4 | Pilot-to-Scale Staircase | p. 127 |
| 5 | Investment AI Maturity Model | p. 128 |

How to Use This Library

For students

Each Slibrary is designed as a mental reference — a "Thumbnail Thinking" collection (Meier, 2025). Use individual sheets when you need to apply a single framework to a problem. Use the full Slibrary when you need to triangulate between several frameworks. The page-anchored hyperlinks in the course website take you directly to the sheet you need.

For practitioners

Treat the 120 sheets as a board-room reference. Each sheet stands alone: a definition you can read aloud, an illustrative figure, and four operational details. The Harvard reference at the foot of each sheet lets you go deeper when needed.

Reading order

There is no required reading order. Browse by Unit (the six course modules) or by Slibrary (the 24 thematic clusters). For first-time readers we suggest starting with Unit 1 · Slibrary 1 (AI Strategy & Governance) and following the cross-references inside each detail modal on the unit pages.

A note on the 9th-section design

The original library design called for nine sections per sheet (definition, who developed it, why it was developed, illustrative figure, three usage columns, innovation vs. predecessor, related frameworks, and sources). This edition uses a streamlined five-section layout that matches the data shown to students in the course-page detail modals. A future revision may restore the full nine-section design.

EU AI Act Risk Pyramid

SLIBRARY 01 · SHEET 1 / 5

Four-tier risk classification from unacceptable to minimal

PROFESSIONAL DEFINITION

A four-tier regulatory classification that categorises AI systems by risk level, from unacceptable (prohibited) to minimal (voluntary codes).

KEY DETAILS

- Unacceptable risk → prohibited applications
- High-risk → conformity assessment required
- Limited risk → transparency obligations
- Minimal risk → voluntary codes of conduct

ILLUSTRATIVE FIGURE



Figure 01.1 — *Four-tier risk classification from unacceptable to minimal*

SLIBRARY 01

AI Strategy & Governance

SOURCE (HARVARD REFERENCING STYLE)

European Parliament (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act). Official Journal of the European Union, L series.

NIST AI RMF Lifecycle

SLIBRARY 01 · SHEET 2 / 5

Continuous risk management: Govern → Map → Measure → Manage

PROFESSIONAL DEFINITION

A voluntary risk management framework comprising four core functions—Govern, Map, Measure, Manage—applied in a continuous cycle to AI systems.

KEY DETAILS

- Four core functions in continuous loop
- Risk-based, not prescriptive approach
- Aligns with ISO and OECD principles
- US voluntary standard for AI governance

ILLUSTRATIVE FIGURE



Figure 01.2 — Continuous risk management: Govern → Map → Measure → Manage

SLIBRARY 01

AI Strategy & Governance

SOURCE (HARVARD REFERENCING STYLE)

National Institute of Standards and Technology (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1. Gaithersburg, MD: NIST.

Perez Techno-Economic S-Curve

SLIBRARY 01 · SHEET 3 / 5

Installation → turning point → deployment phases

PROFESSIONAL DEFINITION

A model showing how technological revolutions unfold in two phases—installation (speculative) and deployment (productive)—separated by a turning point.

KEY DETAILS

- Installation phase driven by speculation
- Turning point: bubble burst and reset
- Deployment phase: production capital dominates
- AI positioned at current turning point

ILLUSTRATIVE FIGURE

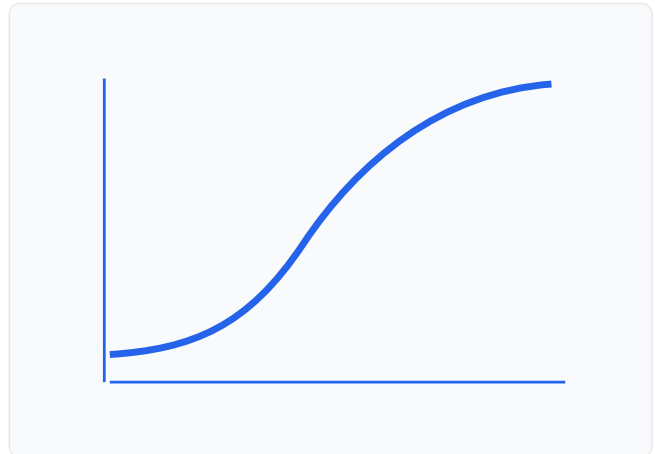


Figure 01.3 — *Installation → turning point → deployment phases*

SLIBRARY 01

AI Strategy & Governance

SOURCE (HARVARD REFERENCING STYLE)

Perez, C. (2002) *Technological Revolutions and Financial Capital: The Dynamics of Bubbles and Golden Ages*. Cheltenham: Edward Elgar.

AI Maturity Staircase

SLIBRARY 01 · SHEET 4 / 5

Four stages from ad hoc to transformational AI adoption

PROFESSIONAL DEFINITION

A four-stage progression model measuring an organisation's AI capability from ad hoc experimentation through to transformational, AI-native operations.

KEY DETAILS

- Ad hoc: isolated experiments
- Experimental: multiple pilots
- Operational: production AI at scale
- Transformational: AI-native organisation

ILLUSTRATIVE FIGURE

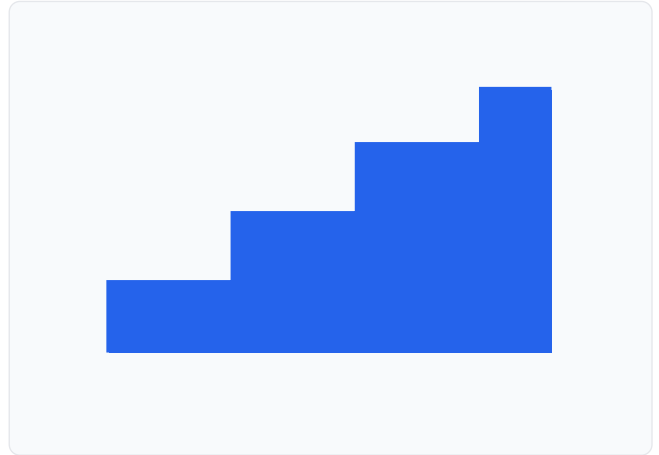


Figure 01.4 — *Four stages from ad hoc to transformational AI adoption*

SLIBRARY 01

AI Strategy & Governance

SOURCE (HARVARD REFERENCING STYLE)

Davenport, T.H. and Harris, J.G. (2017) *Competing on Analytics: Updated, with a New Introduction*. Boston: Harvard Business Review Press.

McKinsey AI High Performers

SLIBRARY 01 · SHEET 5 / 5

3× leadership buy-in drives growth + innovation objectives

PROFESSIONAL DEFINITION

A benchmark comparison showing that AI high-performing organisations are 3× more likely to have strong senior leadership commitment to AI initiatives.

KEY DETAILS

- High performers: 48% strong senior leadership commitment
- 79% pursue innovation as AI objective
- Growth + innovation, not just efficiency
- Source: McKinsey Global Survey 2025

ILLUSTRATIVE FIGURE

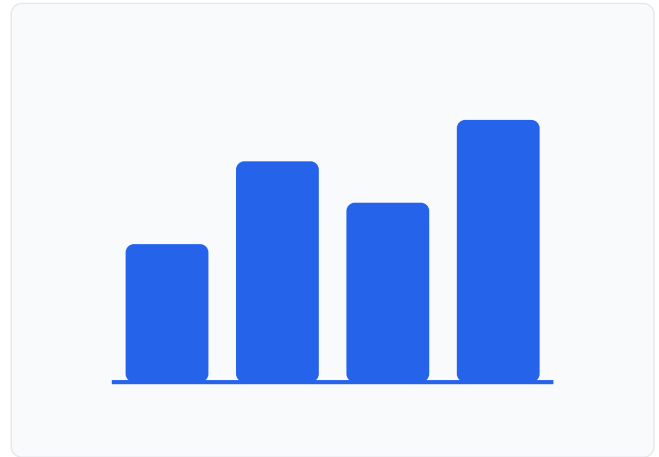


Figure 01.5 — *3× leadership buy-in drives growth + innovation objectives*

SLIBRARY 01

AI Strategy & Governance

SOURCE (HARVARD REFERENCING STYLE)

McKinsey & Company (2025) The state of AI in 2025: Agents, innovation, and transformation. McKinsey Global Survey.

Porter's Five Forces

SLIBRARY 02 · SHEET 1 / 5

Industry structure analysis: rivalry, buyers, suppliers, entrants, substitutes

PROFESSIONAL DEFINITION

An industry analysis framework identifying five competitive forces—rivalry, buyer power, supplier power, threat of entrants, and substitutes—that shape profitability.

KEY DETAILS

- Central rivalry surrounded by four forces
- Maps industry profitability drivers
- AI can shift each force dynamically
- Porter (1979); foundation strategic tool

ILLUSTRATIVE FIGURE

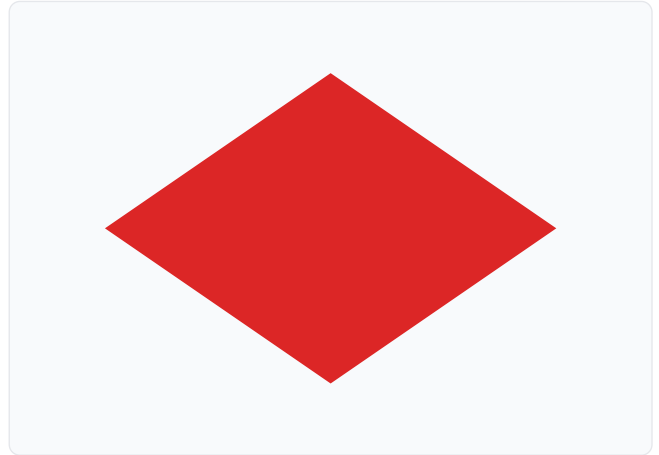


Figure 02.1 — Industry structure analysis: rivalry, buyers, suppliers, entrants, substitutes

SLIBRARY 02

Competitive Advantage

SOURCE (HARVARD REFERENCING STYLE)

Porter, M.E. (1979) 'How competitive forces shape strategy', Harvard Business Review, 57(2), pp. 137–145.

VRIO Framework

SLIBRARY 02 · SHEET 2 / 5

Four-column filter: Valuable → Rare → Inimitable → Organized

PROFESSIONAL DEFINITION

A resource evaluation tool testing whether a firm's assets are Valuable, Rare, costly to Imitate, and supported by Organisation—yielding sustained competitive advantage.

KEY DETAILS

- Sequential test for sustained advantage
- Data and algorithms as rare resources
- Organisation = governance + culture
- Barney (1991); resource-based view

ILLUSTRATIVE FIGURE



Figure 02.2 — *Four-column filter: Valuable → Rare → Inimitable → Organized*

SLIBRARY 02

Competitive Advantage

SOURCE (HARVARD REFERENCING STYLE)

Barney, J.B. (1991) 'Firm resources and sustained competitive advantage', *Journal of Management*, 17(1), pp. 99–120.

Porter's Value Chain

SLIBRARY 02 · SHEET 3 / 5

Primary and support activities creating margin

PROFESSIONAL DEFINITION

A model decomposing a firm into primary and support activities to identify where value is created and where cost advantages or differentiation can be achieved.

KEY DETAILS

- Arrow shape: inputs flow to margin
- Primary: logistics, ops, marketing, service
- Support: infrastructure, HR, tech, procurement
- AI adds value at every activity node

ILLUSTRATIVE FIGURE

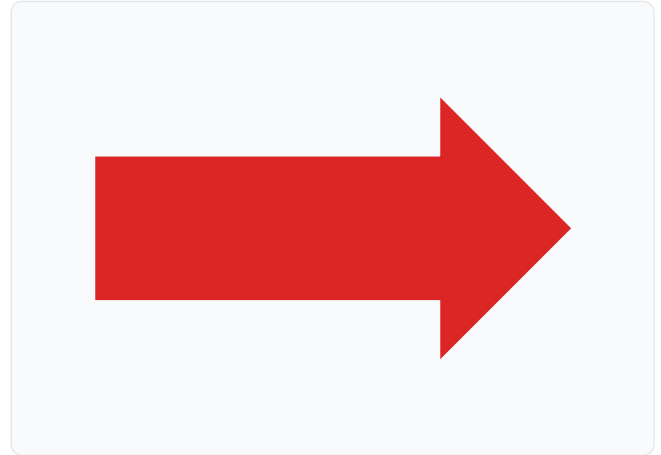


Figure 02.3 — *Primary and support activities creating margin*

SLIBRARY 02

Competitive Advantage

SOURCE (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) *Competitive Advantage: Creating and Sustaining Superior Performance*. New York: Free Press.

Resource-Based View

SLIBRARY 02 · SHEET 4 / 5

Firm resources and capabilities drive competitive heterogeneity

PROFESSIONAL DEFINITION

A theory proposing that firms achieve competitive advantage through resources and capabilities that are heterogeneously distributed and imperfectly mobile across firms.

KEY DETAILS

- Heterogeneity: firms differ in resources
- Immobility: resources hard to transfer
- Data moats as modern RBV resources
- Wernerfelt (1984); Barney (1991)

ILLUSTRATIVE FIGURE



Figure 02.4 — *Firm resources and capabilities drive competitive heterogeneity*

SLIBRARY 02

Competitive Advantage

SOURCE (HARVARD REFERENCING STYLE)

Wernerfelt, B. (1984) 'A resource-based view of the firm', Strategic Management Journal, 5(2), pp. 171–180.

Disruption S-Curves

SLIBRARY 02 · SHEET 5 / 5

Incumbent vs. entrant performance trajectories over time

PROFESSIONAL DEFINITION

A model of innovation trajectories where a new technology's performance curve eventually crosses and surpasses the incumbent's, signalling market disruption.

KEY DETAILS

- Two intersecting S-curves
- Crossing point signals disruption
- Entrants start below market needs
- Christensen (1997) innovation theory

ILLUSTRATIVE FIGURE

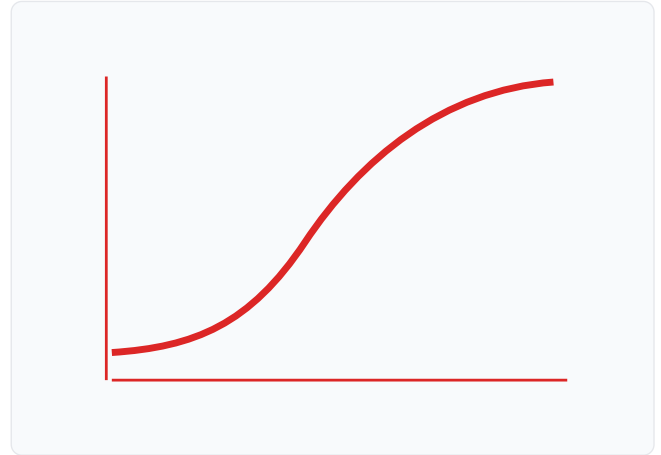


Figure 02.5 — Incumbent vs. entrant performance trajectories over time

SLIBRARY 02

Competitive Advantage

SOURCE (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail*. Boston: Harvard Business School Press.

CRISP-DM Cycle

SLIBRARY 03 · SHEET 1 / 5

Six-phase circular process for data mining and ML projects

PROFESSIONAL DEFINITION

A six-phase, iterative methodology for data mining and ML projects: business understanding, data understanding, data preparation, modelling, evaluation, and deployment.

KEY DETAILS

- Business Understanding → Data Understanding
- Data Preparation → Modelling
- Evaluation → Deployment
- Industry standard since 1999

ILLUSTRATIVE FIGURE

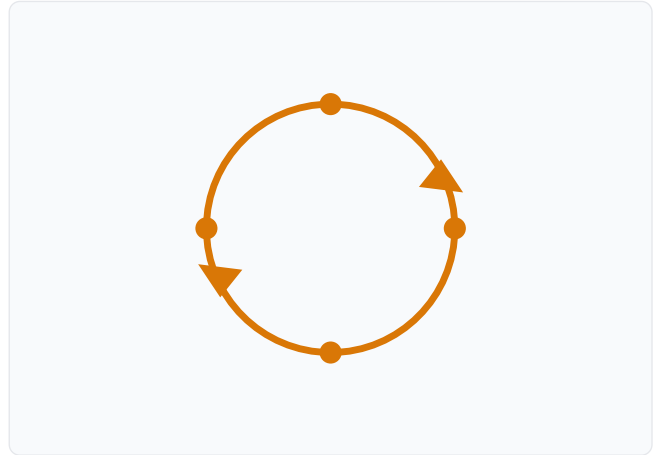


Figure 03.1 — Six-phase circular process for data mining and ML projects

SLIBRARY 03

AI Project Lifecycle

SOURCE (HARVARD REFERENCING STYLE)

Wirth, R. and Hipp, J. (2000) 'CRISP-DM: Towards a standard process model for data mining', Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, pp. 29–39.

DDED Stage-Gate

SLIBRARY 03 · SHEET 2 / 5

Design → Develop → Evaluate → Deploy with go/no-go gates

PROFESSIONAL DEFINITION

A four-phase AI project framework—Design, Develop, Evaluate, Deploy—with go/no-go decision gates between each phase to ensure business alignment.

KEY DETAILS

- Four sequential phases
- Go/no-go decision at each gate
- Aligns AI with business case
- MARTI301 core framework

ILLUSTRATIVE FIGURE

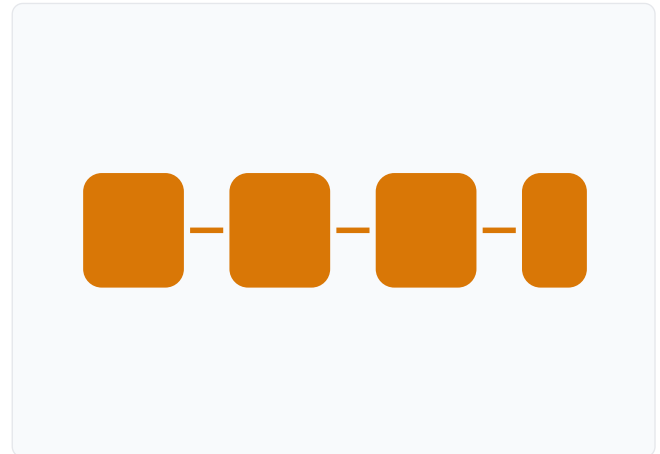


Figure 03.2 — *Design → Develop → Evaluate → Deploy with go/no-go gates*

SLIBRARY 03

AI Project Lifecycle

SOURCE (HARVARD REFERENCING STYLE)

Cooper, R.G. (2017) *Winning at New Products: Creating Value Through Innovation*. 5th edn. New York: Basic Books. Adapted for AI.

Build vs. Buy 2×2

SLIBRARY 03 · SHEET 3 / 5

Strategic value × capability matrix for AI sourcing decisions

PROFESSIONAL DEFINITION

A decision matrix plotting strategic value against internal capability to determine whether an AI solution should be built in-house, purchased, or partnered.

KEY DETAILS

- High value + high capability = build
- Low value + commodity = buy/partner
- Guides resource allocation
- Links to talent strategy (Unit 4)

ILLUSTRATIVE FIGURE

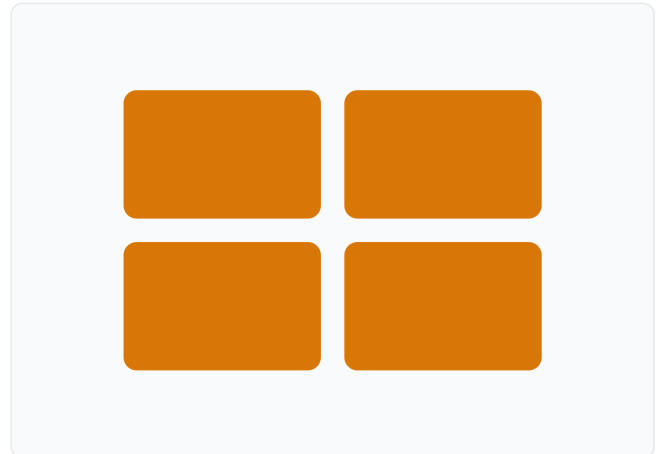


Figure 03.3 — *Strategic value × capability matrix for AI sourcing decisions*

SLIBRARY 03

AI Project Lifecycle

SOURCE (HARVARD REFERENCING STYLE)

Iansiti, M. and Lakhani, K.R. (2020) Competing in the Age of AI. Boston: Harvard Business Review Press.

AI Failure Funnel

SLIBRARY 03 · SHEET 4 / 5

87% of AI projects never reach production—narrowing at each stage

PROFESSIONAL DEFINITION

An empirical observation that approximately 87% of AI projects never reach production, with attrition occurring at data, talent, and integration barriers.

KEY DETAILS

- Wide top: many ideas and pilots
- Narrows at data, talent, integration barriers
- Only 13% reach production deployment
- Gartner / VentureBeat research data

ILLUSTRATIVE FIGURE

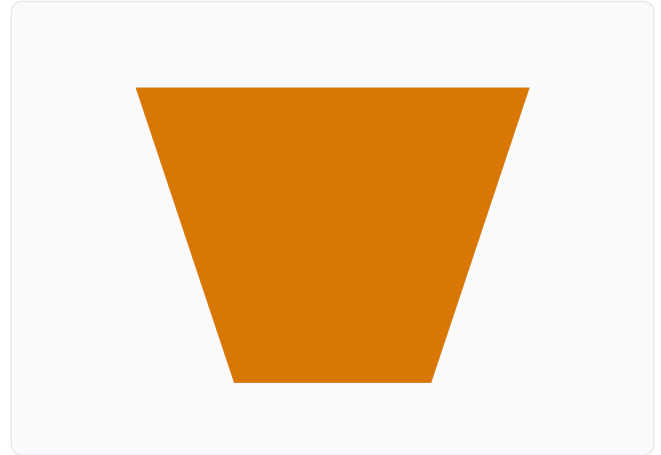


Figure 03.4 — 87% of AI projects never reach production—narrowing at each stage

SLIBRARY 03

AI Project Lifecycle

SOURCE (HARVARD REFERENCING STYLE)

VentureBeat (2019) 'Why do 87% of data science projects never make it into production?', VentureBeat, 19 July.

Data Readiness Ladder

SLIBRARY 03 · SHEET 5 / 5

Band D (hearsay) through Band A (production-grade) data quality

PROFESSIONAL DEFINITION

A four-band classification of data maturity from Band D (hearsay, unverifiable) through Band A (production-grade, continuously maintained).

KEY DETAILS

- Band D: hearsay, unverifiable
- Band C: cleaned and structured
- Band B: task-ready for ML
- Band A: production-grade, maintained

ILLUSTRATIVE FIGURE

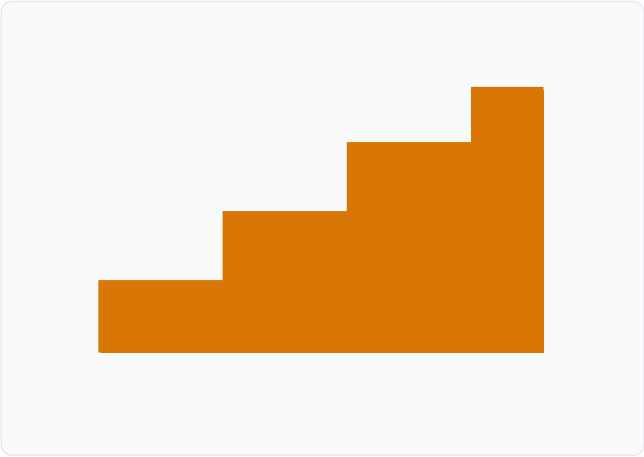


Figure 03.5 — *Band D (hearsay) through Band A (production-grade) data quality*

SLIBRARY 03

AI Project Lifecycle

SOURCE (HARVARD REFERENCING STYLE)

Lawrence, N.D. (2017) 'Data readiness levels', arXiv preprint arXiv:1705.02245.

Prediction-Judgment Split

SLIBRARY 04 · SHEET 1 / 5

AI = prediction engine, humans = judgment + action

PROFESSIONAL DEFINITION

A model positing that AI radically reduces the cost of prediction while the value of human judgment—deciding what to do with predictions—correspondingly increases.

KEY DETAILS

- AI drops the cost of prediction
- Humans provide judgment on actions
- Complementary, not substitutive roles
- Agrawal, Gans & Goldfarb (2018)

ILLUSTRATIVE FIGURE

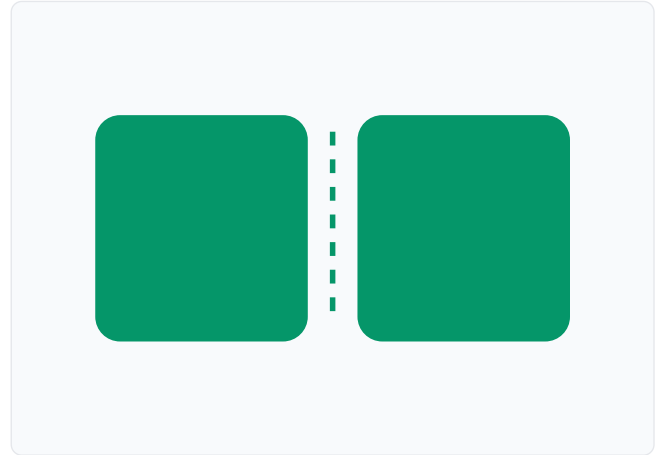


Figure 04.1 — *AI = prediction engine, humans = judgment + action*

SLIBRARY 04

Human-AI Collaboration

SOURCE (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines: The Simple Economics of Artificial Intelligence. Boston: Harvard Business Review Press.

Centaur Model

SLIBRARY 04 · SHEET 2 / 5

Human + AI outperforms either alone—originated in chess

PROFESSIONAL DEFINITION

A human-AI teaming concept, originating from chess, where the combined performance of human strategic oversight and AI computational power exceeds either alone.

KEY DETAILS

- Kasparov's insight after Deep Blue
- Strategic oversight + AI computational speed
- Best results from genuine collaboration
- Applicable across all business domains

ILLUSTRATIVE FIGURE

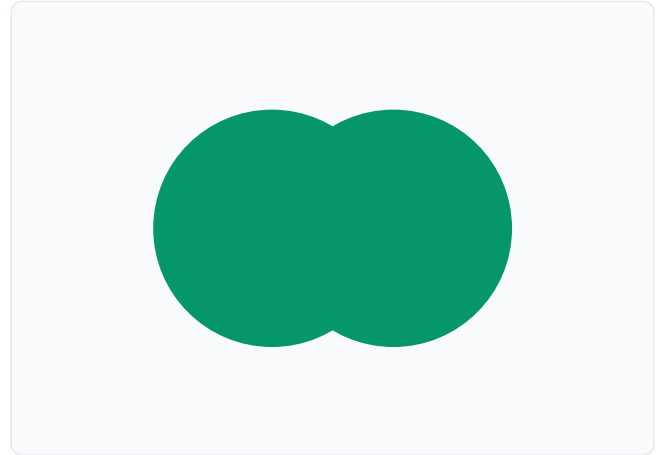


Figure 04.2 — *Human + AI outperforms either alone—originated in chess*

SLIBRARY 04

Human-AI Collaboration

SOURCE (HARVARD REFERENCING STYLE)

Kasparov, G. (2017) Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins. London: John Murray.

Automation-Augmentation Spectrum

SLIBRARY 04 · SHEET 3 / 5

Full automation ← → *full augmentation for task-level analysis*

PROFESSIONAL DEFINITION

A continuum for analysing work at the task level, ranging from full automation (AI replaces the task) to full augmentation (AI enhances human performance).

KEY DETAILS

- Most roles sit in the middle
- Task-level, not job-level analysis
- AI handles routine; humans handle exceptions
- Frey & Osborne framework reframed

ILLUSTRATIVE FIGURE

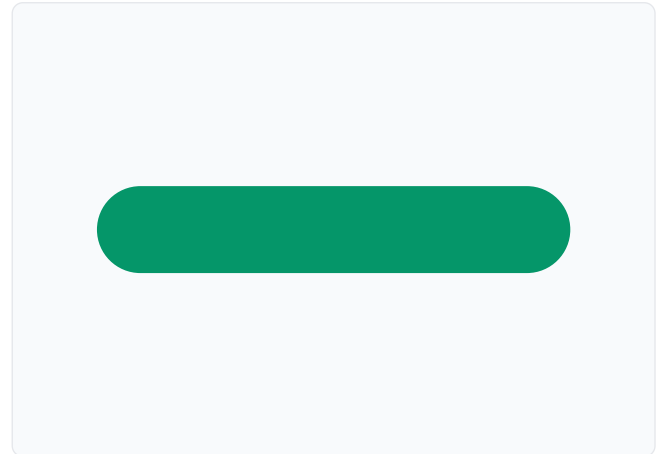


Figure 04.3 — *Full automation* ← → *full augmentation for task-level analysis*

SLIBRARY 04

Human-AI Collaboration

SOURCE (HARVARD REFERENCING STYLE)

Raisch, S. and Krakowski, S. (2021) 'Artificial intelligence and management: The automation-augmentation paradox', *Academy of Management Review*, 46(1), pp. 192–210.

BCG People Investment

SLIBRARY 04 · SHEET 4 / 5

High performers invest 2× in upskilling over replacing

PROFESSIONAL DEFINITION

Research finding that AI high-performing organisations invest approximately twice as much in workforce upskilling as in workforce replacement.

KEY DETAILS

- Reskilling > replacing for AI adoption
- Culture + capability = successful adoption
- Investment in change management critical
- BCG / McKinsey (2025) research

ILLUSTRATIVE FIGURE

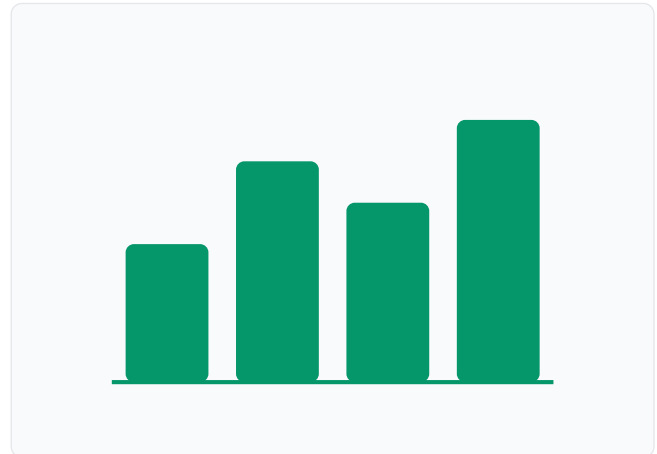


Figure 04.4 — High performers invest 2× in upskilling over replacing

SLIBRARY 04

Human-AI Collaboration

SOURCE (HARVARD REFERENCING STYLE)

Ransbotham, S. et al. (2020) 'Expanding AI's impact with organizational learning', MIT Sloan Management Review and BCG.

Human-in-the-Loop Feedback

SLIBRARY 04 · SHEET 5 / 5

AI output → human review → correction → continuous learning

PROFESSIONAL DEFINITION

A system design pattern in which AI outputs are reviewed, corrected, and approved by humans, creating a continuous learning loop that improves model accuracy and trust.

KEY DETAILS

- Circular feedback between AI and human
- Critical for high-stakes domains
- Reduces bias and increases trust
- Foundation of responsible AI deployment

ILLUSTRATIVE FIGURE

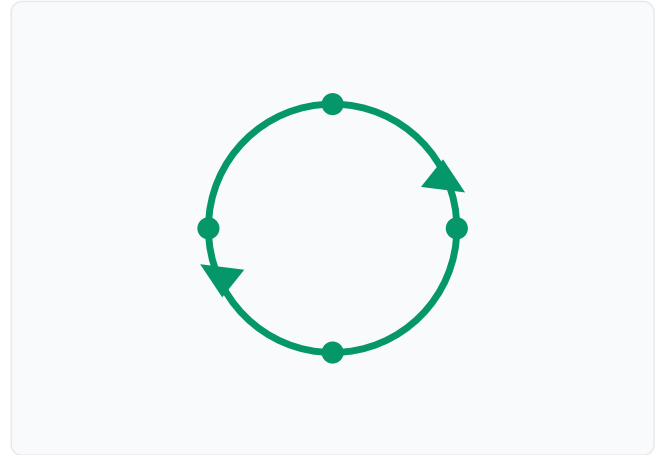


Figure 04.5 — *AI output → human review → correction → continuous learning*

SLIBRARY 04

Human-AI Collaboration

SOURCE (HARVARD REFERENCING STYLE)

Amershi, S. et al. (2019) 'Guidelines for human-AI interaction', Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, pp. 1–13.

ML Learning Types Taxonomy

SLIBRARY 05 · SHEET 1 / 5

Supervised, unsupervised, and reinforcement learning branches

PROFESSIONAL DEFINITION

A classification of machine learning into three paradigms: supervised (labelled data), unsupervised (pattern discovery), and reinforcement (reward-based) learning.

KEY DETAILS

- Supervised: labelled data → prediction
- Unsupervised: patterns in unlabelled data
- Reinforcement: reward-based learning
- Tree diagram with three main branches

ILLUSTRATIVE FIGURE

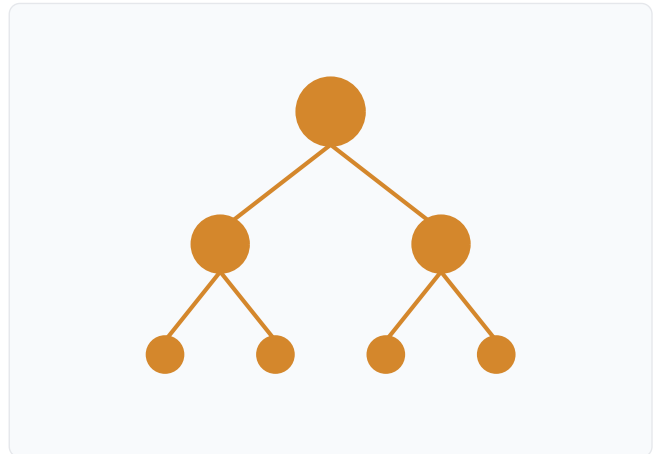


Figure 05.1 — Supervised, unsupervised, and reinforcement learning branches

SLIBRARY 05

ML Models & Techniques

SOURCE (HARVARD REFERENCING STYLE)

Bishop, C.M. (2006) Pattern Recognition and Machine Learning. New York: Springer.

Bias-Variance Tradeoff

SLIBRARY 05 · SHEET 2 / 5

U-shaped total error curve with sweet spot in the middle

PROFESSIONAL DEFINITION

The principle that model prediction error comprises reducible bias (underfitting) and variance (overfitting), with total error minimised at their balance point.

KEY DETAILS

- Low complexity = high bias (underfitting)
- High complexity = high variance (overfitting)
- Sweet spot minimises total error
- Core ML model selection concept

ILLUSTRATIVE FIGURE

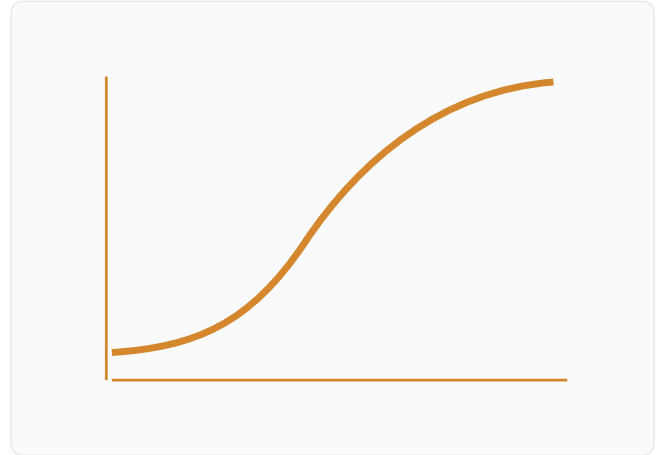


Figure 05.2 — *U-shaped total error curve with sweet spot in the middle*

SLIBRARY 05

ML Models & Techniques

SOURCE (HARVARD REFERENCING STYLE)

Hastie, T., Tibshirani, R. and Friedman, J. (2009) The Elements of Statistical Learning. 2nd edn. New York: Springer.

Confusion Matrix 2×2

SLIBRARY 05 · SHEET 3 / 5

TP/FP/FN/TN grid—precision, recall, F1 derive from it

PROFESSIONAL DEFINITION

A tabular evaluation of classifier performance showing true positives, false positives, false negatives, and true negatives, from which precision, recall, and F1 are derived.

KEY DETAILS

- True Positives / False Positives
- False Negatives / True Negatives
- Precision, recall, F1 score derived
- Core model evaluation thumbnail

ILLUSTRATIVE FIGURE

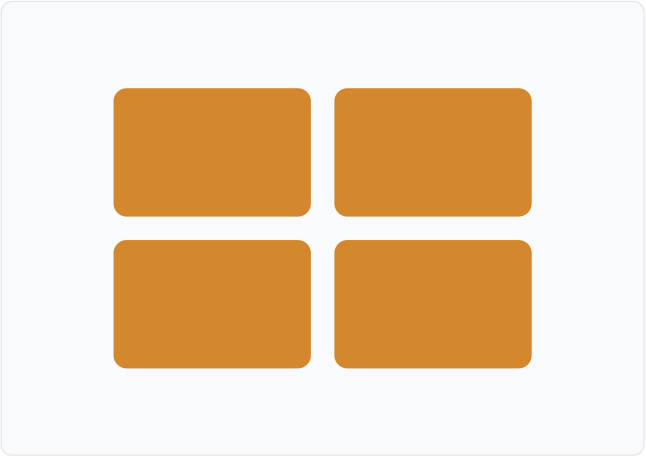


Figure 05.3 — TP/FP/FN/TN grid—precision, recall, F1 derive from it

SLIBRARY 05

ML Models & Techniques

SOURCE (HARVARD REFERENCING STYLE)

Fawcett, T. (2006) 'An introduction to ROC analysis', Pattern Recognition Letters, 27(8), pp. 861–874.

Feature Importance Chart

SLIBRARY 05 · SHEET 4 / 5

Horizontal bars ranked by weight—explainability for stakeholders

PROFESSIONAL DEFINITION

A ranked visualisation of input variables by their contribution to model predictions, enabling explainability and stakeholder communication.

KEY DETAILS

- Which inputs drive the model most?
- Key for explainability (XAI)
- Communicates ML to business leaders
- SHAP values or permutation importance

ILLUSTRATIVE FIGURE

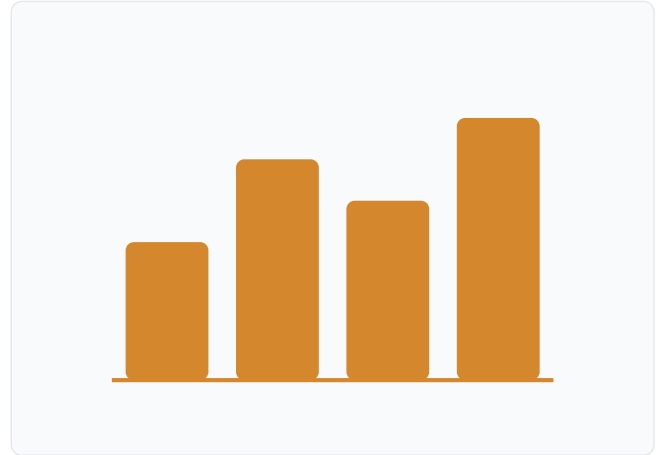


Figure 05.4 — Horizontal bars ranked by weight—explainability for stakeholders

SLIBRARY 05

ML Models & Techniques

SOURCE (HARVARD REFERENCING STYLE)

Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', Advances in Neural Information Processing Systems, 30.

Model Selection Flowchart

SLIBRARY 05 · SHEET 5 / 5

Decision tree: structured → regression/classification/clustering

PROFESSIONAL DEFINITION

A decision tree guiding practitioners from data type and problem type to the appropriate machine learning algorithm family.

KEY DETAILS

- Start: structured or unstructured data?
- Branches by problem type
- Guides non-technical stakeholders
- Scikit-learn-style decision tree

ILLUSTRATIVE FIGURE

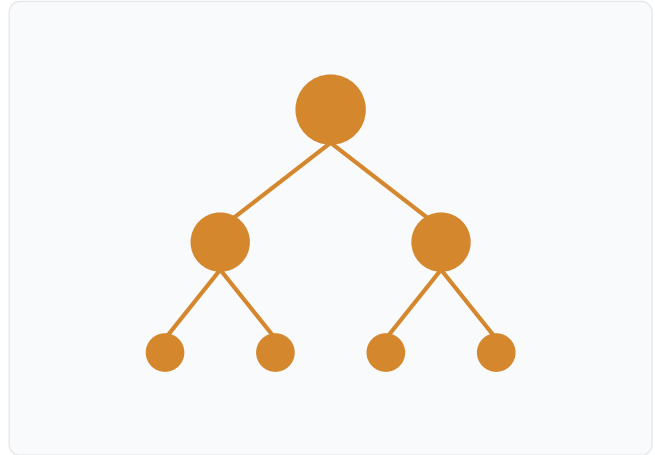


Figure 05.5 — *Decision tree: structured → regression/classification/clustering*

SLIBRARY 05

ML Models & Techniques

SOURCE (HARVARD REFERENCING STYLE)

Scikit-learn (2023) 'Choosing the right estimator', scikit-learn.org.

OODA Loop

SLIBRARY 06 · SHEET 1 / 5

Observe → Orient → Decide → Act—speed of cycle = advantage

PROFESSIONAL DEFINITION

A decision-cycle model—Observe, Orient, Decide, Act—where competitive advantage derives from cycling through decisions faster than opponents.

KEY DETAILS

- Speed of decision cycle = competitive edge
- Agility in volatile markets
- AI accelerates Observe and Orient phases
- Boyd (1976); O'Callaghan (2023)

ILLUSTRATIVE FIGURE



Figure 06.1 — *Observe → Orient → Decide → Act—speed of cycle = advantage*

SLIBRARY 06

Decision Intelligence

SOURCE (HARVARD REFERENCING STYLE)

Boyd, J.R. (1976) Destruction and Creation. Cited in: O'Callaghan, M. (2023) Decision Intelligence. Boca Raton: CRC Press.

Weighted Decision Matrix

SLIBRARY 06 · SHEET 2 / 5

Criteria × weight scoring grid for transparent trade-offs

PROFESSIONAL DEFINITION

A structured evaluation tool that scores options against weighted criteria, quantifying subjective trade-offs into a transparent, auditable decision.

KEY DETAILS

- Quantifies subjective trade-offs
- AI can automate the scoring layer
- Transparent and auditable decisions
- Standard decision analysis tool

ILLUSTRATIVE FIGURE

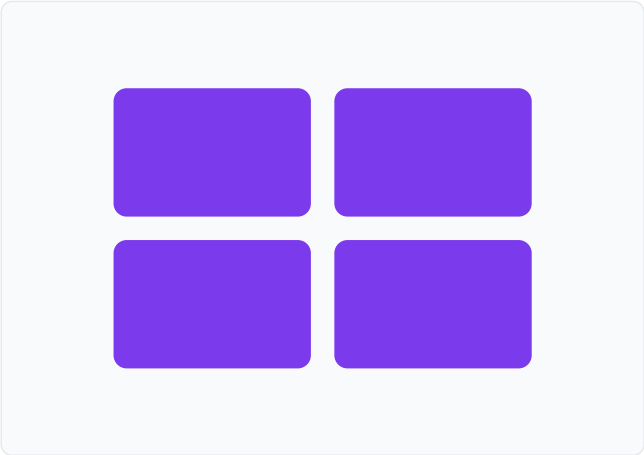


Figure 06.2 — Criteria × weight scoring grid for transparent trade-offs

SLIBRARY 06

Decision Intelligence

SOURCE (HARVARD REFERENCING STYLE)

O'Callaghan, M. (2023) Decision Intelligence: Human–Machine Integration for Decision-Making. Boca Raton: CRC Press.

Cynefin Framework

SLIBRARY 06 · SHEET 3 / 5

Simple → Complicated → Complex → Chaotic decision contexts

PROFESSIONAL DEFINITION

A sense-making model categorising decision contexts as simple, complicated, complex, or chaotic—each requiring a different leadership approach.

KEY DETAILS

- Match decision approach to context
- AI excels in complicated domain
- Complex domain needs experimentation
- Snowden & Boone (2007)

ILLUSTRATIVE FIGURE

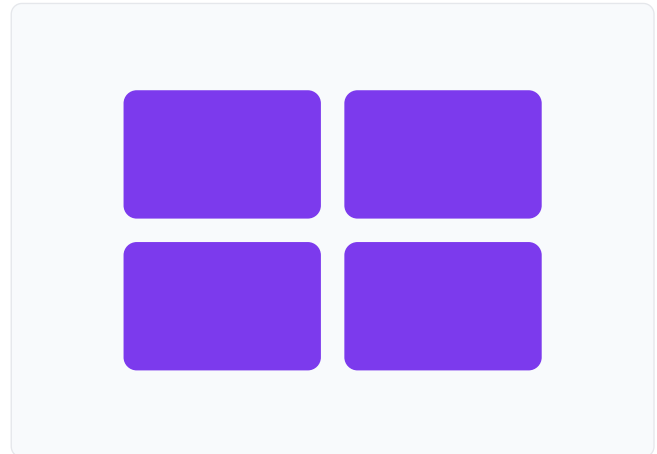


Figure 06.3 — *Simple → Complicated → Complex → Chaotic decision contexts*

SLIBRARY 06

Decision Intelligence

SOURCE (HARVARD REFERENCING STYLE)

Snowden, D.J. and Boone, M.E. (2007) 'A leader's framework for decision making', Harvard Business Review, 85(11), pp. 68–76.

IDSS Architecture Layers

SLIBRARY 06 · SHEET 4 / 5

Data → Analytics → UI layers with human judgment on top

PROFESSIONAL DEFINITION

A three-layer architecture for Intelligent Decision Support Systems: data layer, analytics/AI layer, and user interface, with human judgment at the apex.

KEY DETAILS

- Three-layer architecture stack
- Human judgment sits above all layers
- AI augments, doesn't replace decisions
- Core Unit 2 framework

ILLUSTRATIVE FIGURE

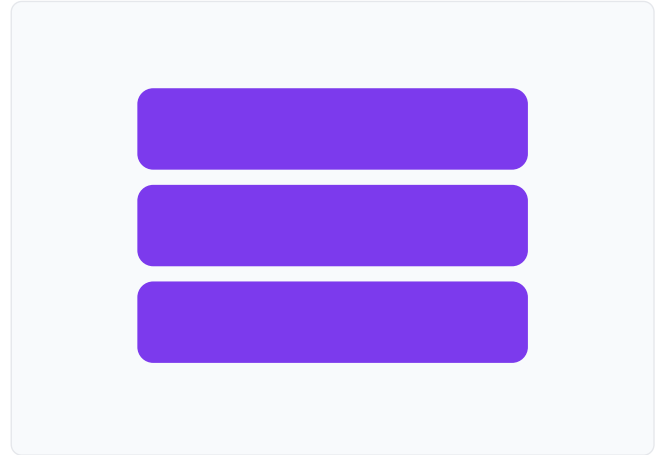


Figure 06.4 — *Data → Analytics → UI layers with human judgment on top*

SLIBRARY 06

Decision Intelligence

SOURCE (HARVARD REFERENCING STYLE)

Phillips-Wren, G. et al. (2021) 'Intelligent decision support systems in the era of big data', Knowledge-Based Systems, 227, 107222.

Prediction Machines Model

SLIBRARY 06 · SHEET 5 / 5

AI drops cost of prediction—judgment remains human domain

PROFESSIONAL DEFINITION

An economic framework arguing that AI's primary function is to reduce the cost of prediction, thereby increasing the value of complements: judgment, data, and action.

KEY DETAILS

- Prediction $\neq$ decision
- Cheaper prediction raises value of judgment
- Complements and data become more valuable
- Agrawal, Gans & Goldfarb (2018)

ILLUSTRATIVE FIGURE

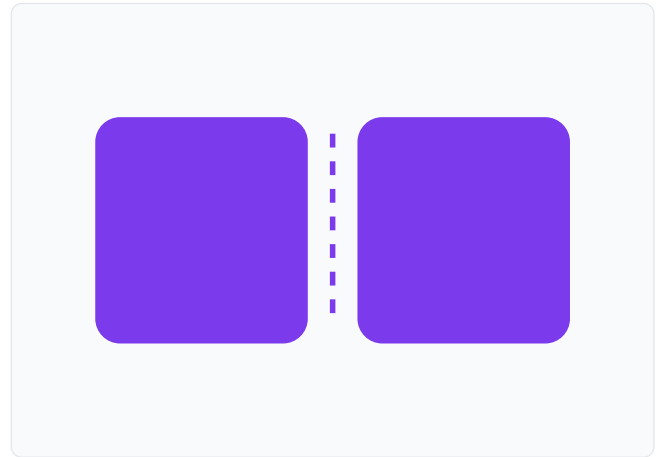


Figure 06.5 — *AI drops cost of prediction—judgment remains human domain*

SLIBRARY 06

Decision Intelligence

SOURCE (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines: The Simple Economics of Artificial Intelligence. Boston: Harvard Business Review Press.

Data Value Chain

SLIBRARY 07 · SHEET 1 / 5

Collect → Store → Process → Analyse → Act with value increasing

PROFESSIONAL DEFINITION

A linear model showing how raw data is transformed through collection, storage, processing, analysis, and action stages into business value.

KEY DETAILS

- Linear arrow: value increases left to right
- Where leakage occurs at each step
- AI accelerates processing and analysis
- Foundation for data monetisation

ILLUSTRATIVE FIGURE

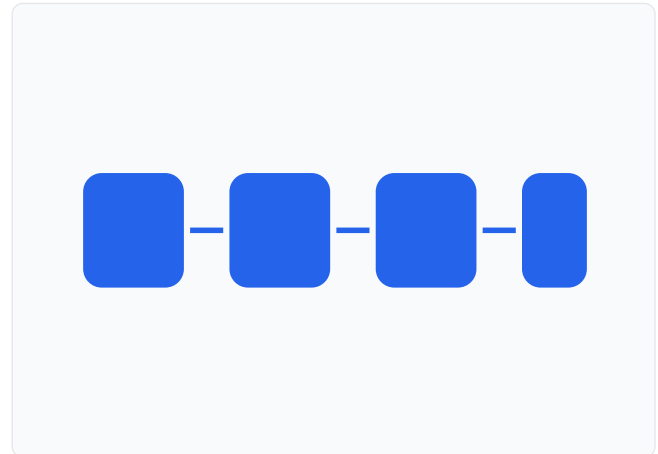


Figure 07.1 — *Collect → Store → Process → Analyse → Act with value increasing*

SLIBRARY 07

Data Strategy

SOURCE (HARVARD REFERENCING STYLE)

Porter, M.E. and Millar, V.E. (1985) 'How information gives you competitive advantage', Harvard Business Review, 63(4), pp. 149–160. Adapted for data.

DIKW Pyramid

SLIBRARY 07 · SHEET 2 / 5

Data → Information → Knowledge → Wisdom hierarchy

PROFESSIONAL DEFINITION

A knowledge hierarchy arranging Data, Information, Knowledge, and Wisdom in ascending layers—volume decreases upward while insight and value increase.

KEY DETAILS

- Volume decreases upward
- AI accelerates D → I → K layers
- Wisdom still requires human judgment
- Ackoff (1989); classic knowledge mgmt

ILLUSTRATIVE FIGURE



Figure 07.2 — *Data → Information → Knowledge → Wisdom hierarchy*

SLIBRARY 07

Data Strategy

SOURCE (HARVARD REFERENCING STYLE)

Ackoff, R.L. (1989) 'From data to wisdom', *Journal of Applied Systems Analysis*, 16(1), pp. 3–9.

Data Flywheel

SLIBRARY 07 · SHEET 3 / 5

More data → better model → better product → more users → more data

PROFESSIONAL DEFINITION

A self-reinforcing virtuous cycle: more data improves models, better models improve products, better products attract users, more users generate more data.

KEY DETAILS

- Circular virtuous cycle
- Self-reinforcing competitive advantage
- Amazon, Netflix, Google exemplars
- Explains winner-take-most dynamics

ILLUSTRATIVE FIGURE



Figure 07.3 — *More data → better model → better product → more users → more data*

SLIBRARY 07

Data Strategy

SOURCE (HARVARD REFERENCING STYLE)

Ng, A. (2018) 'AI is the new electricity', Stanford University HAI.

Data Quality Dimensions

SLIBRARY 07 · SHEET 4 / 5

Accuracy, completeness, consistency, timeliness, validity, uniqueness

PROFESSIONAL DEFINITION

Six measurable attributes of data fitness: accuracy, completeness, consistency, timeliness, validity, and uniqueness—collectively determining data's fitness for purpose.

KEY DETAILS

- Six dimensions in hexagonal shape
- Poor quality = garbage in, garbage out
- Each dimension independently measurable
- Foundation for data governance

ILLUSTRATIVE FIGURE



Figure 07.4 — Accuracy, completeness, consistency, timeliness, validity, uniqueness

SLIBRARY 07

Data Strategy

SOURCE (HARVARD REFERENCING STYLE)

Wang, R.Y. and Strong, D.M. (1996) 'Beyond accuracy: What data quality means to data consumers', *Journal of Management Information Systems*, 12(4), pp. 5–33.

ETL Pipeline Diagram

SLIBRARY 07 · SHEET 5 / 5

Extract → Transform → Load from sources to data warehouse

PROFESSIONAL DEFINITION

A data engineering workflow—Extract, Transform, Load—that moves data from heterogeneous sources into a structured analytical warehouse.

KEY DETAILS

- Sources on left, warehouse on right
- Where AI automates transformation
- Modern variant: ELT in cloud
- Foundational data engineering visual

ILLUSTRATIVE FIGURE

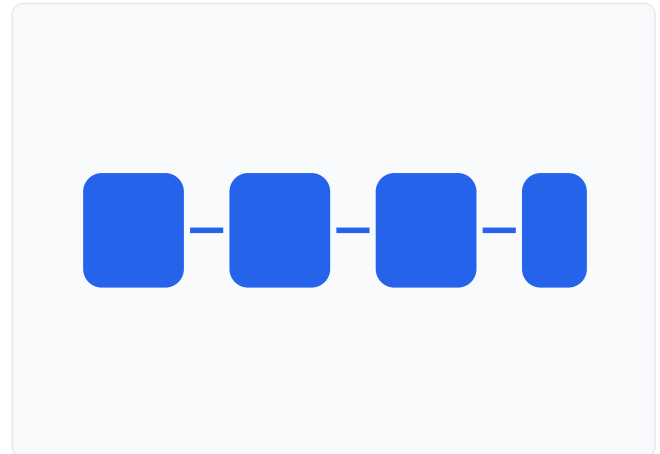


Figure 07.5 — *Extract → Transform → Load from sources to data warehouse*

SLIBRARY 07

Data Strategy

SOURCE (HARVARD REFERENCING STYLE)

Kimball, R. and Ross, M. (2013) The Data Warehouse Toolkit. 3rd edn. Indianapolis: Wiley.

AI Marketing Funnel

SLIBRARY 08 · SHEET 1 / 5

Awareness → Consideration → Conversion with AI at every stage

PROFESSIONAL DEFINITION

An enhanced marketing funnel where AI automates or augments each stage—from predictive lead scoring at awareness through dynamic pricing at conversion.

KEY DETAILS

- AI touchpoints at every funnel stage
- Predictive lead scoring at top
- Dynamic pricing at conversion
- Gentsch (2019); Kotler et al.

ILLUSTRATIVE FIGURE



Figure 08.1 — Awareness → Consideration → Conversion with AI at every stage

SLIBRARY 08

AI-Driven Marketing

SOURCE (HARVARD REFERENCING STYLE)

Gentsch, P. (2019) AI in Marketing, Sales and Service. Cham: Palgrave Macmillan.

RFM Segmentation Matrix

SLIBRARY 08 · SHEET 2 / 5

Recency × Frequency × Monetary value customer clustering

PROFESSIONAL DEFINITION

A customer segmentation method clustering buyers by Recency, Frequency, and Monetary value to identify high-value segments for targeted engagement.

KEY DETAILS

- 3D cube or heatmap thumbnail
- Identifies high-value customer segments
- Classic + ML-enhanced clustering
- Foundation for personalisation

ILLUSTRATIVE FIGURE

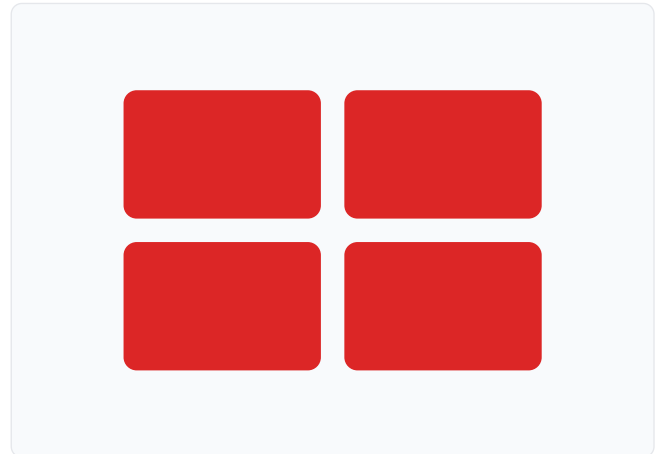


Figure 08.2 — *Recency × Frequency × Monetary value customer clustering*

SLIBRARY 08

AI-Driven Marketing

SOURCE (HARVARD REFERENCING STYLE)

Fader, P.S., Hardie, B.G.S. and Lee, K.L. (2005) 'RFM and CLV: Using iso-value curves for customer base analysis', *Journal of Marketing Research*, 42(4), pp. 415–430.

A/B Testing Framework

SLIBRARY 08 · SHEET 3 / 5

Control vs. treatment with statistical significance threshold

PROFESSIONAL DEFINITION

An experimental design comparing a control group against one or more treatment variations to measure the causal impact of changes on key metrics.

KEY DETAILS

- Split diagram: control vs. treatment
- AI enables multi-armed bandits
- Continuous optimisation cycle
- Beyond simple A/B to multivariate

ILLUSTRATIVE FIGURE

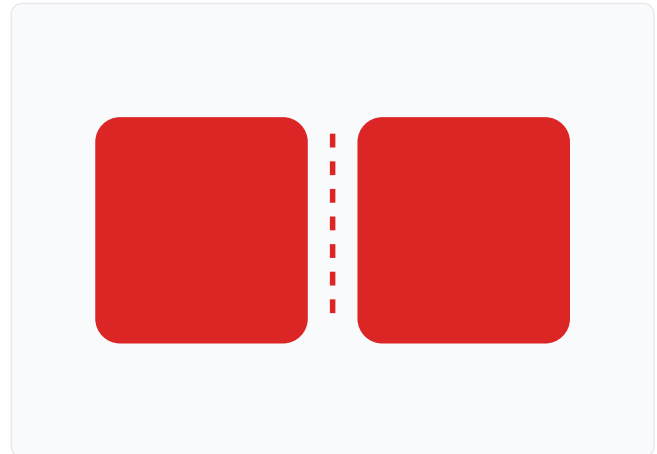


Figure 08.3 — *Control vs. treatment with statistical significance threshold*

SLIBRARY 08

AI-Driven Marketing

SOURCE (HARVARD REFERENCING STYLE)

Kohavi, R., Tang, D. and Xu, Y. (2020) Trustworthy Online Controlled Experiments. Cambridge: Cambridge University Press.

Recommendation Engine

SLIBRARY 08 · SHEET 4 / 5

Collaborative + content-based filtering architecture

PROFESSIONAL DEFINITION

An algorithmic system combining collaborative filtering (user behaviour patterns) and content-based filtering to predict user preferences and personalise offerings.

KEY DETAILS

- User-item interaction matrix
- Collaborative + content-based hybrid
- Drives 35% of Amazon revenue
- Netflix Prize catalysed the field

ILLUSTRATIVE FIGURE

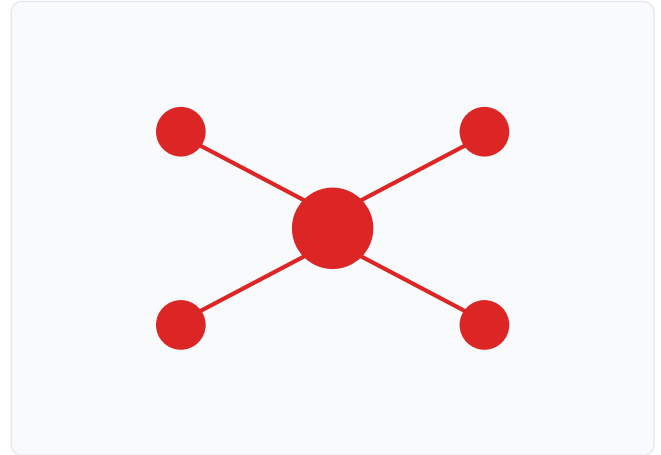


Figure 08.4 — Collaborative + content-based filtering architecture

SLIBRARY 08

AI-Driven Marketing

SOURCE (HARVARD REFERENCING STYLE)

Ricci, F. et al. (eds.) (2015) Recommender Systems Handbook. 2nd edn. New York: Springer.

Customer Lifetime Value

SLIBRARY 08 · SHEET 5 / 5

Revenue × margin × retention / discount rate over customer lifespan

PROFESSIONAL DEFINITION

The present value of all future cash flows attributed to a customer relationship, calculated as revenue × margin × retention rate, discounted over time.

KEY DETAILS

- Curve: value over customer lifespan
- AI predicts churn to protect CLV
- Links acquisition cost to long-term value
- Bridge to Unit 3 CX metrics

ILLUSTRATIVE FIGURE

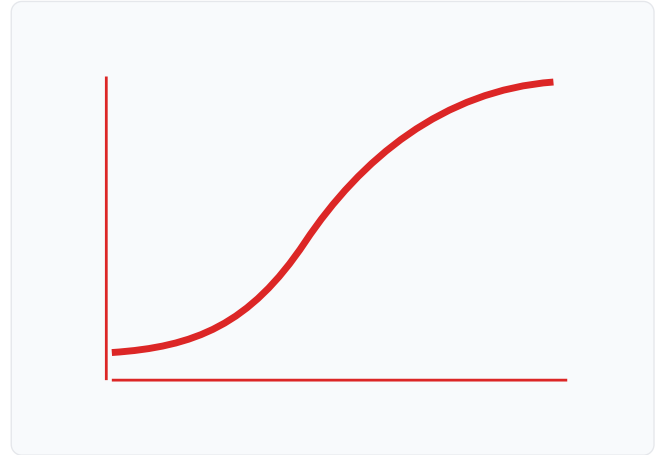


Figure 08.5 — *Revenue × margin × retention / discount rate over customer lifespan*

SLIBRARY 08

AI-Driven Marketing

SOURCE (HARVARD REFERENCING STYLE)

Gupta, S. and Lehmann, D.R. (2005) Managing Customers as Investments. Upper Saddle River: Wharton School Publishing.

Omnichannel Journey Map

SLIBRARY 09 · SHEET 1 / 5

Horizontal timeline with channels stacked and AI touchpoints

PROFESSIONAL DEFINITION

A visualisation tracing the customer's path across all channels from awareness through advocacy, identifying AI intervention points.

KEY DETAILS

- Awareness → consideration → purchase → advocacy
- Channels: web, app, store, call centre
- AI touchpoints marked at each stage
- Identifies gaps and friction points

ILLUSTRATIVE FIGURE

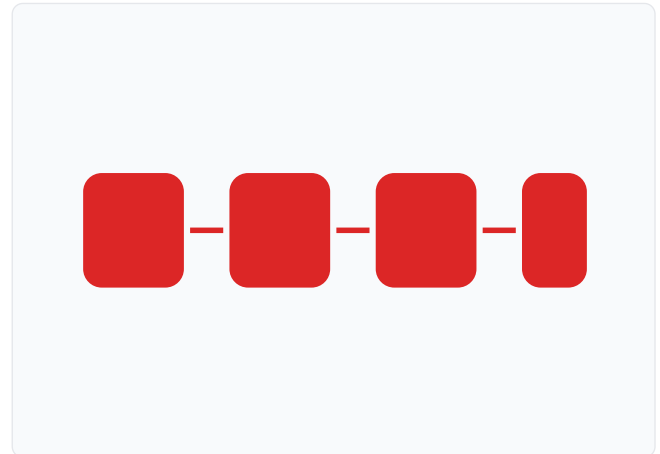


Figure 09.1 — Horizontal timeline with channels stacked and AI touchpoints

SLIBRARY 09

Customer Journey & CX

SOURCE (HARVARD REFERENCING STYLE)

Lemon, K.N. and Verhoef, P.C. (2016) 'Understanding customer experience throughout the customer journey', *Journal of Marketing*, 80(6), pp. 69–96.

CX Experience Pyramid

SLIBRARY 09 · SHEET 2 / 5

Functional → accessible → enjoyable—AI lifts each layer

PROFESSIONAL DEFINITION

A three-tier model of customer needs: functional (meets needs), accessible (easy to use), and emotional (enjoyable)—with AI enhancing each layer.

KEY DETAILS

- Base: meet needs (functional)
- Middle: easy to use (accessible)
- Top: enjoyable (emotional)
- AI differentiates at each layer

ILLUSTRATIVE FIGURE

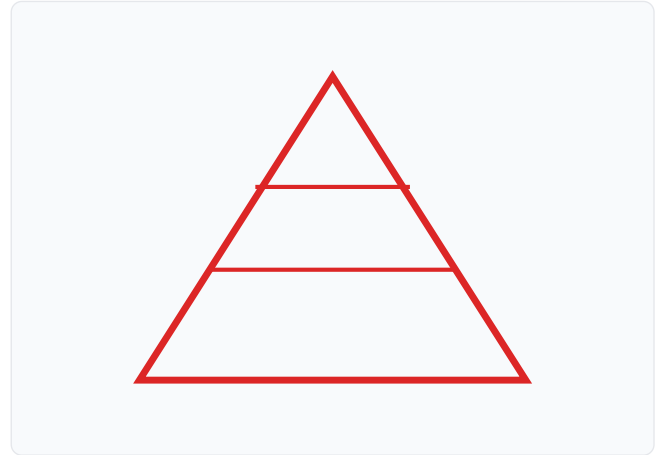


Figure 09.2 — *Functional → accessible → enjoyable—AI lifts each layer*

SLIBRARY 09

Customer Journey & CX

SOURCE (HARVARD REFERENCING STYLE)

Forrester Research (2016) The Customer Experience Pyramid. Forrester Research Inc.

Service Blueprint

SLIBRARY 09 · SHEET 3 / 5

Frontstage vs. backstage swim lanes with line of visibility

PROFESSIONAL DEFINITION

A process diagram separating frontstage (customer-visible) from backstage (internal) actions with a line of visibility, mapping all support processes.

KEY DETAILS

- Frontstage: customer-facing actions
- Line of visibility separates front/back
- Backstage: support processes
- Shostack (1984); updated for AI era

ILLUSTRATIVE FIGURE

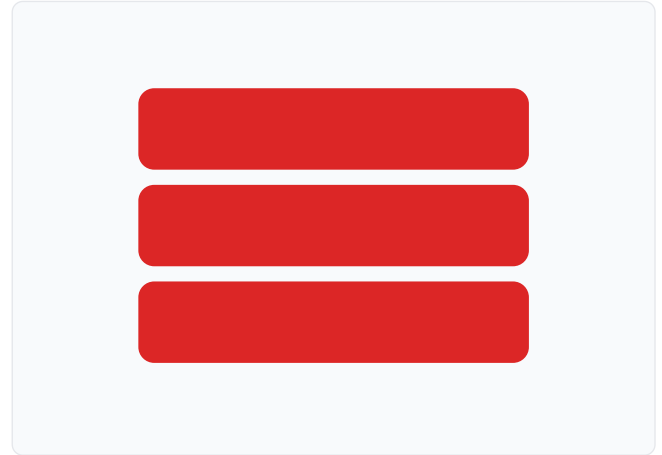


Figure 09.3 — *Frontstage vs. backstage swim lanes with line of visibility*

SLIBRARY 09

Customer Journey & CX

SOURCE (HARVARD REFERENCING STYLE)

Shostack, G.L. (1984) 'Designing services that deliver', Harvard Business Review, 62(1), pp. 133–139.

Touchpoint Matrix

Channel × journey stage heatmap showing interaction intensity

SLIBRARY 09 · SHEET 4 / 5

PROFESSIONAL DEFINITION

A grid mapping customer journey stages against channels, with heatmap shading to reveal over-served and under-served interaction areas.

KEY DETAILS

- Reveals over-served and under-served areas
- Heatmap colour coding by intensity
- Prioritises where AI investment goes
- Data-driven channel optimisation

ILLUSTRATIVE FIGURE

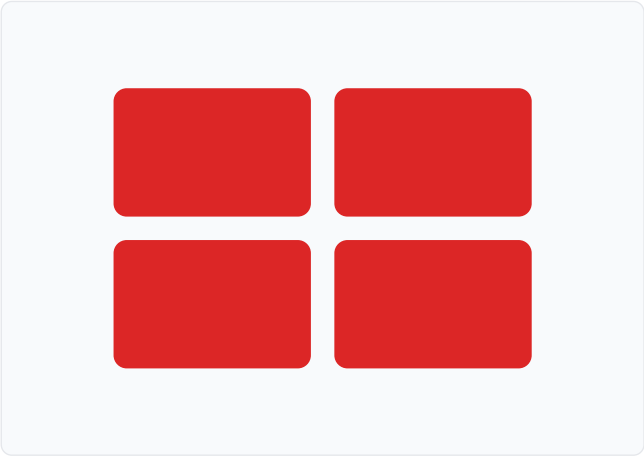


Figure 09.4 — Channel × journey stage heatmap showing interaction intensity

SLIBRARY 09

Customer Journey & CX

SOURCE (HARVARD REFERENCING STYLE)

Rawson, A., Duncan, E. and Jones, C. (2013) 'The truth about customer experience', Harvard Business Review, 91(9), pp. 90–98.

Moments of Truth Timeline

SLIBRARY 09 · SHEET 5 / 5

Zero → First → Second → Ultimate MOT along customer journey

PROFESSIONAL DEFINITION

A framework identifying four critical customer decision points: Zero, First, Second, and Ultimate moments of truth.

KEY DETAILS

- Zero MOT: online research (Google)
- First MOT: shelf or landing page
- Second MOT: usage experience
- Ultimate MOT: sharing and advocacy

ILLUSTRATIVE FIGURE

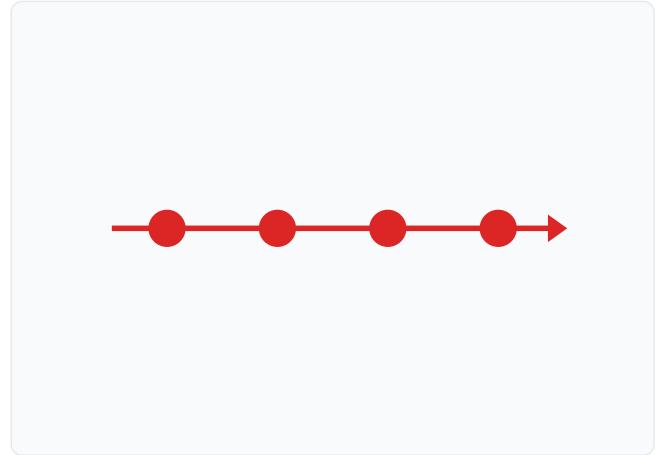


Figure 09.5 — Zero → First → Second → Ultimate MOT along customer journey

SLIBRARY 09

Customer Journey & CX

SOURCE (HARVARD REFERENCING STYLE)

Lecinski, J. (2011) Winning the Zero Moment of Truth. Google Inc.

NLP Pipeline Diagram

SLIBRARY 10 · SHEET 1 / 5

Input → Tokenise → Parse → Understand → Respond with feedback

PROFESSIONAL DEFINITION

A sequential processing architecture: input → tokenisation → parsing → semantic understanding → response generation, with feedback for continuous improvement.

KEY DETAILS

- Linear flow with feedback loop
- Where LLMs transformed each step
- Foundation for chatbot architecture
- Pre-LLM vs. post-LLM comparison

ILLUSTRATIVE FIGURE

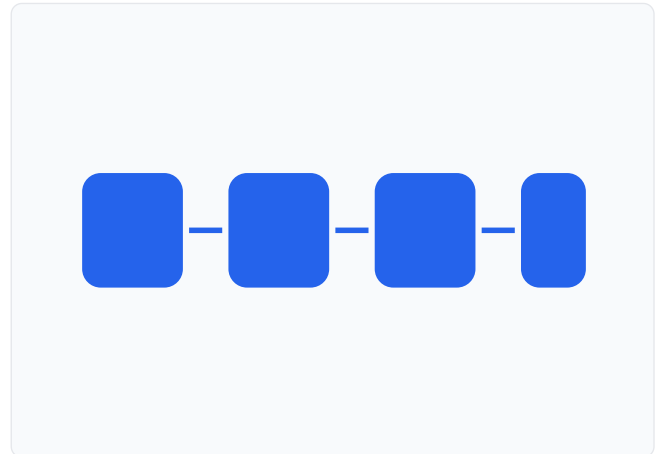


Figure 10.1 — *Input → Tokenise → Parse → Understand → Respond with feedback*

SLIBRARY 10

Conversational AI & NLP

SOURCE (HARVARD REFERENCING STYLE)

Jurafsky, D. and Martin, J.H. (2023) Speech and Language Processing. 3rd edn. draft. Stanford University.

Intent-Entity-Context Model

SLIBRARY 10 · SHEET 2 / 5

What does the user want? What parameters? What state?

PROFESSIONAL DEFINITION

A natural language understanding framework decomposing user utterances into intent (goal), entities (parameters), and context (conversation state).

KEY DETAILS

- Intent: user's goal
- Entity: key parameters
- Context: conversation state
- Core NLU triangle diagram

ILLUSTRATIVE FIGURE



Figure 10.2 — *What does the user want? What parameters? What state?*

SLIBRARY 10

Conversational AI & NLP

SOURCE (HARVARD REFERENCING STYLE)

Jurafsky, D. and Martin, J.H. (2023) Speech and Language Processing. 3rd edn. draft. Stanford University.

Chatbot Decision Tree

SLIBRARY 10 · SHEET 3 / 5

Branching flowchart from greeting with fallback to human

PROFESSIONAL DEFINITION

A branching dialogue architecture where intent classification nodes route conversations to responses, actions, or human escalation fallback paths.

KEY DETAILS

- Decision nodes: intent classification
- Leaf nodes: responses or actions
- Fallback paths to human agent
- Designed for graceful degradation

ILLUSTRATIVE FIGURE

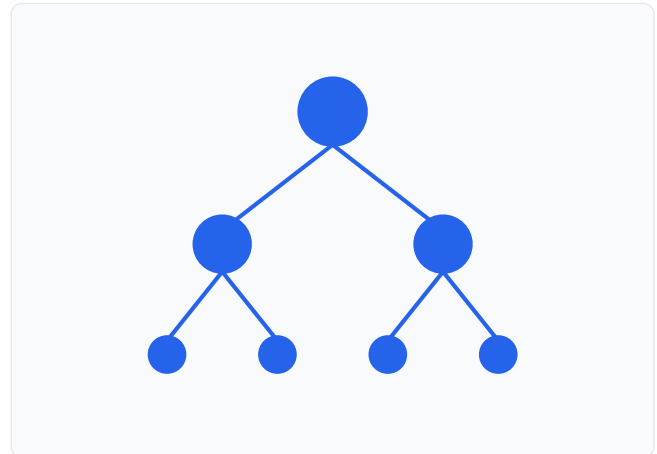


Figure 10.3 — *Branching flowchart from greeting with fallback to human*

SLIBRARY 10

Conversational AI & NLP

SOURCE (HARVARD REFERENCING STYLE)

McTear, M. (2020) Conversational AI: Dialogue Systems, Conversational Agents, and Chatbots. San Rafael: Morgan & Claypool.

Sentiment Analysis Spectrum

SLIBRARY 10 · SHEET 4 / 5

Negative → neutral → positive real-time mood tracking

PROFESSIONAL DEFINITION

A real-time measurement scale mapping customer language from negative through neutral to positive, with threshold-based alerts triggering interventions.

KEY DETAILS

- Linear bar with threshold markers
- Real-time customer mood tracking
- Triggers alerts at negative thresholds
- Feeds into CX dashboards

ILLUSTRATIVE FIGURE

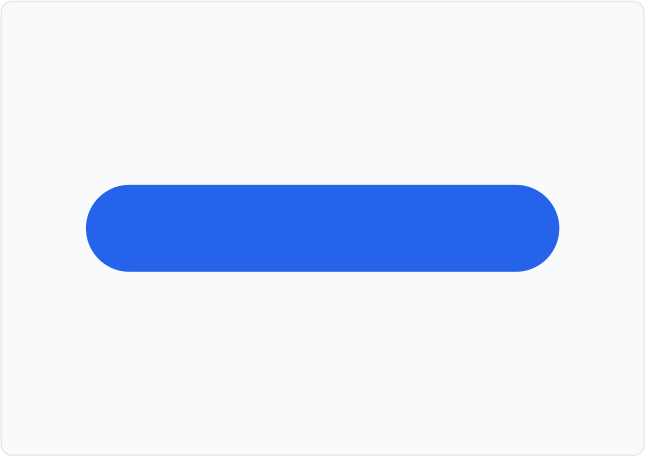


Figure 10.4 — Negative → neutral → positive real-time mood tracking

SLIBRARY 10

Conversational AI & NLP

SOURCE (HARVARD REFERENCING STYLE)

Liu, B. (2020) Sentiment Analysis: Mining Opinions, Sentiments, and Emotions. 2nd edn. Cambridge: Cambridge University Press.

Human Escalation Flowchart

SLIBRARY 10 · SHEET 5 / 5

AI handles L1 → confidence check → human handoff with context

PROFESSIONAL DEFINITION

A confidence-gated handoff process where AI handles initial enquiries and transfers to human agents—with full context—when confidence falls below threshold.

KEY DETAILS

- Below confidence threshold → handoff
- Context packet transfers with escalation
- Balances efficiency with empathy
- Key for high-stakes interactions

ILLUSTRATIVE FIGURE

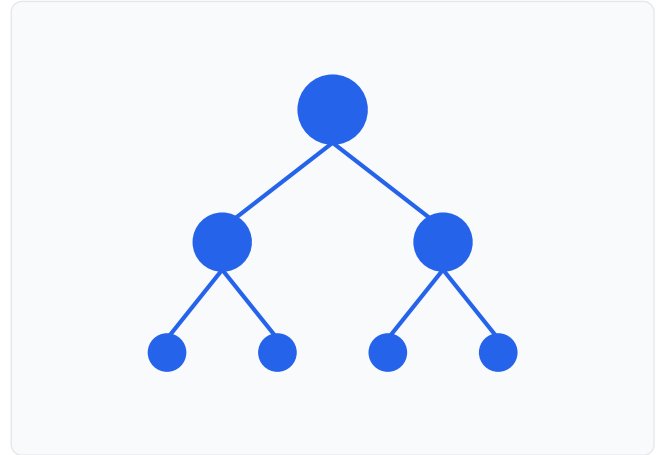


Figure 10.5 — *AI handles L1 → confidence check → human handoff with context*

SLIBRARY 10

Conversational AI & NLP

SOURCE (HARVARD REFERENCING STYLE)

McTear, M. (2020) Conversational AI: Dialogue Systems, Conversational Agents, and Chatbots. San Rafael: Morgan & Claypool.

NPS-CSAT-CES Triangle

SLIBRARY 11 · SHEET 1 / 5

Three complementary lenses on customer experience measurement

PROFESSIONAL DEFINITION

Three complementary CX metrics: Net Promoter Score (loyalty), Customer Satisfaction (transaction), and Customer Effort Score (ease).

KEY DETAILS

- NPS: loyalty and advocacy (top)
- CSAT: transaction satisfaction (left)
- CES: customer effort score (right)
- Use all three for complete picture

ILLUSTRATIVE FIGURE



Figure 11.1 — *Three complementary lenses on customer experience measurement*

SLIBRARY 11

CX Metrics & Business Impact

SOURCE (HARVARD REFERENCING STYLE)

Reichheld, F.F. (2003) 'The one number you need to grow', Harvard Business Review, 81(12), pp. 46–54.

Churn Waterfall Chart

SLIBRARY 11 · SHEET 2 / 5

Starting customers → lost → retained with attrition drivers

PROFESSIONAL DEFINITION

A cascading bar chart visualising customer attrition from starting base through specific loss drivers to retained count, quantifying revenue at risk.

KEY DETAILS

- Waterfall bars show attrition drivers
- AI predicts at-risk customer segments
- Quantifies revenue at stake
- Prioritises retention interventions

ILLUSTRATIVE FIGURE

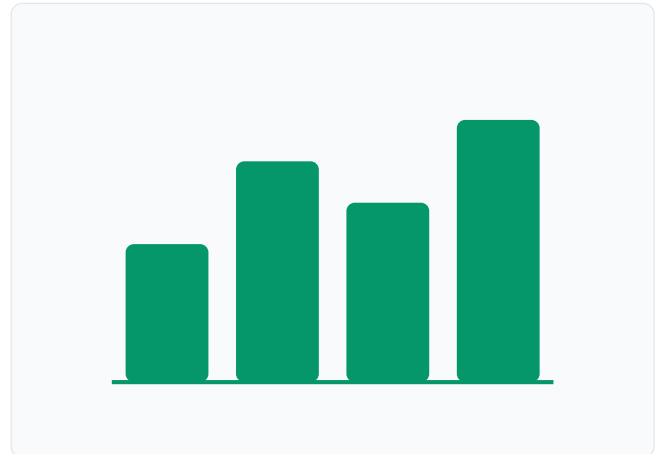


Figure 11.2 — *Starting customers → lost → retained with attrition drivers*

SLIBRARY 11

CX Metrics & Business Impact

SOURCE (HARVARD REFERENCING STYLE)

Ascarza, E. (2018) 'Retention futility: Targeting high-risk customers is not effective', *Journal of Marketing Research*, 55(1), pp. 80–98.

CLV-to-CAC Ratio

SLIBRARY 11 · SHEET 3 / 5

Lifetime value vs. acquisition cost—healthy ratio > 3:1

PROFESSIONAL DEFINITION

A unit economics diagnostic comparing Customer Lifetime Value to Customer Acquisition Cost; a healthy ratio exceeds 3:1.

KEY DETAILS

- Side-by-side comparison bars
- Healthy business ratio > 3:1
- AI optimises both sides simultaneously
- Unit economics diagnostic thumbnail

ILLUSTRATIVE FIGURE

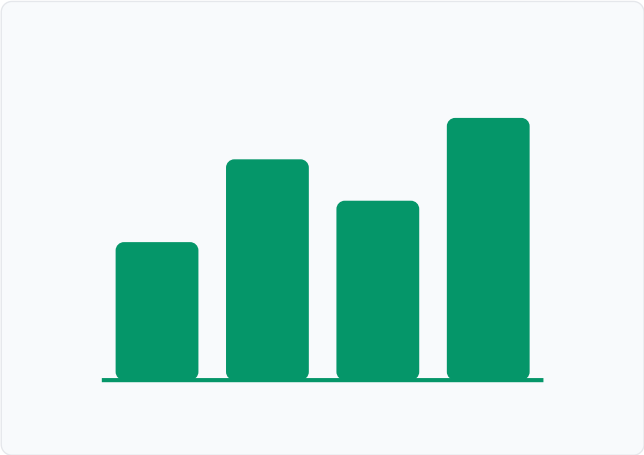


Figure 11.3 — Lifetime value vs. acquisition cost—healthy ratio > 3:1

SLIBRARY 11

CX Metrics & Business Impact

SOURCE (HARVARD REFERENCING STYLE)

Gupta, S. and Lehmann, D.R. (2005) Managing Customers as Investments. Upper Saddle River: Wharton School Publishing.

Loyalty Loop Diagram

SLIBRARY 11 · SHEET 4 / 5

Consider → buy → enjoy → bond replacing the linear funnel

PROFESSIONAL DEFINITION

A circular decision journey model—consider, buy, enjoy, bond—replacing the linear funnel for repeat customers.

KEY DETAILS

- Circular replaces linear funnel
- AI shortens re-entry to the loop
- Focus on post-purchase experience
- McKinsey consumer decision journey

ILLUSTRATIVE FIGURE

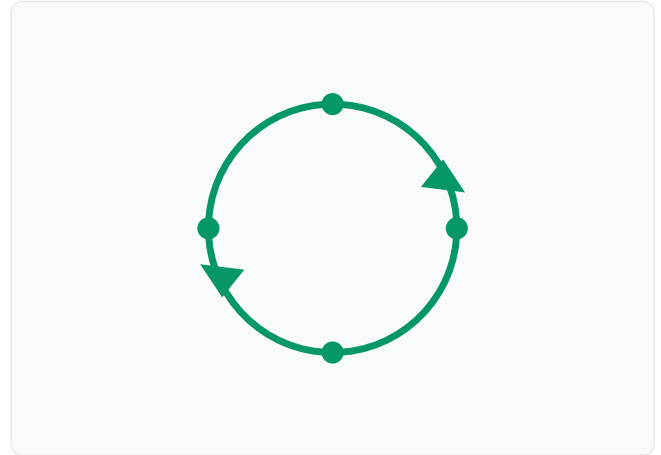


Figure 11.4 — *Consider → buy → enjoy → bond replacing the linear funnel*

SLIBRARY 11

CX Metrics & Business Impact

SOURCE (HARVARD REFERENCING STYLE)

Court, D. et al. (2009) 'The consumer decision journey', McKinsey Quarterly, June.

CX-to-Revenue Bridge

SLIBRARY 11 · SHEET 5 / 5

CX metric → intermediate outcomes → revenue/EBIT impact

PROFESSIONAL DEFINITION

A translation model linking CX improvement metrics through intermediate outcomes to financial results.

KEY DETAILS

- Left: CX improvement metric
- Middle: churn, AOV, frequency outcomes
- Right: revenue and EBIT impact
- Builds the business case for CX AI

ILLUSTRATIVE FIGURE

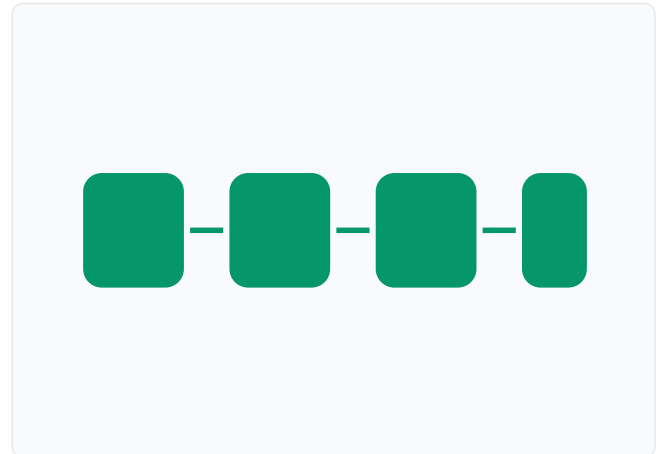


Figure 11.5 — *CX metric → intermediate outcomes → revenue/EBIT impact*

SLIBRARY 11

CX Metrics & Business Impact

SOURCE (HARVARD REFERENCING STYLE)

Program on Customer Strategy, Columbia Business School. Adapted for MARTI301.

Marketing Evolution Timeline

SLIBRARY 12 · SHEET 1 / 5

1.0 Product → 2.0 Customer → ... → 6.0 Immersive

PROFESSIONAL DEFINITION

A six-era historical progression from product-centric marketing (1.0) to immersive marketing (6.0).

KEY DETAILS

- Six eras of marketing evolution
- 6.0: immersive and multisensory
- AI + AR/VR + IoT convergence
- Kotler, Kartajaya & Setiawan

ILLUSTRATIVE FIGURE

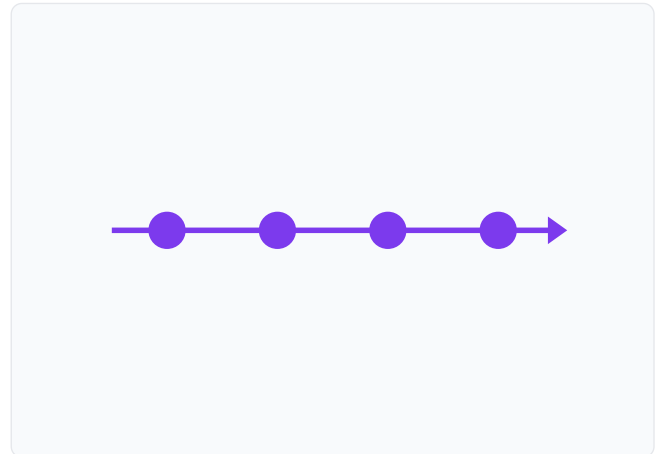


Figure 12.1 — *1.0 Product → 2.0 Customer → ... → 6.0 Immersive*

SLIBRARY 12

Marketing 6.0 & Immersive

SOURCE (HARVARD REFERENCING STYLE)

Kotler, P., Kartajaya, H. and Setiawan, I. (2023) Marketing 6.0: The Future Is Immersive. Hoboken: Wiley.

Phygital Experience Model

SLIBRARY 12 · SHEET 2 / 5

Physical + digital = phygital—seamless overlap zone

PROFESSIONAL DEFINITION

A Venn diagram concept where physical and digital experiences overlap to create seamless 'phygital' interactions, bridged by AI.

KEY DETAILS

- Venn diagram: physical $\cap$ digital
- AI bridges the two worlds
- Nike, IKEA, Sephora exemplars
- Where the experience magic happens

ILLUSTRATIVE FIGURE



Figure 12.2 — *Physical + digital = phygital—seamless overlap zone*

SLIBRARY 12

Marketing 6.0 & Immersive

SOURCE (HARVARD REFERENCING STYLE)

Kotler, P., Kartajaya, H. and Setiawan, I. (2023) Marketing 6.0: The Future Is Immersive. Hoboken: Wiley.

Metaverse CX Framework

SLIBRARY 12 · SHEET 3 / 5

Hardware → platform → experience layers with avatars

PROFESSIONAL DEFINITION

A three-layer architecture for metaverse customer experiences: hardware, platform, and experience layer.

KEY DETAILS

- Three architectural layers
- Customer avatars interact with brand
- Digital twins of products and spaces
- Early-stage but high strategic potential

ILLUSTRATIVE FIGURE

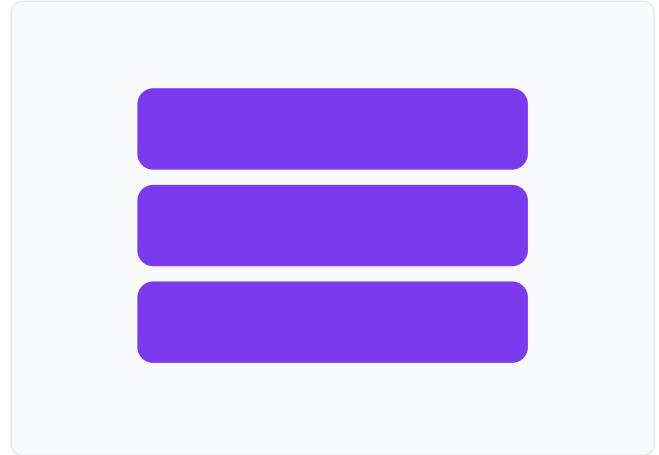


Figure 12.3 — *Hardware → platform → experience layers with avatars*

SLIBRARY 12

Marketing 6.0 & Immersive

SOURCE (HARVARD REFERENCING STYLE)

Ball, M. (2022) The Metaverse: And How It Will Revolutionize Everything. New York: Liveright.

Spatial Computing Stack

SLIBRARY 12 · SHEET 4 / 5

AR/VR/MR/XR taxonomy with sensor → processing → display

PROFESSIONAL DEFINITION

A technology taxonomy (AR, VR, MR, XR) with a three-layer architecture: sensors, processors, displays.

KEY DETAILS

- Technology taxonomy clarification
- Sensors → processing → display layers
- AI enables real-time scene understanding
- Apple Vision Pro, Meta Quest exemplars

ILLUSTRATIVE FIGURE

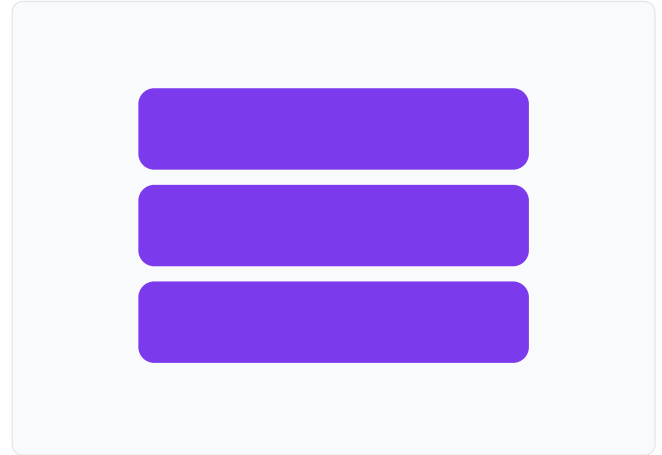


Figure 12.4 — AR/VR/MR/XR taxonomy with sensor → processing → display

SLIBRARY 12

Marketing 6.0 & Immersive

SOURCE (HARVARD REFERENCING STYLE)

Greenwold, S. (2003) Spatial Computing. MSc thesis, MIT.

Kotler 6.0 Ecosystem Map

SLIBRARY 12 · SHEET 5 / 5

Five technologies centred on immersive customer value

PROFESSIONAL DEFINITION

An interconnected system of five enabling technologies—AI, AR, VR, IoT, blockchain—centred on creating immersive customer value.

KEY DETAILS

- AI, AR, VR, IoT, blockchain
- Interconnected ecosystem diagram
- Centred on immersive customer value
- Kotler et al. (2023) framework

ILLUSTRATIVE FIGURE

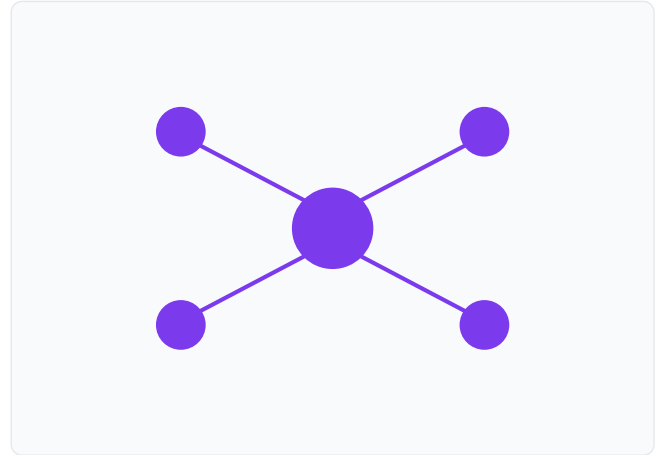


Figure 12.5 — Five technologies centred on immersive customer value

SLIBRARY 12

Marketing 6.0 & Immersive

SOURCE (HARVARD REFERENCING STYLE)

Kotler, P., Kartajaya, H. and Setiawan, I. (2023) Marketing 6.0: The Future Is Immersive. Hoboken: Wiley.

FinTech Ecosystem Map

SLIBRARY 13 · SHEET 1 / 5

Concentric circles: banks → fintechs → big tech with AI core

PROFESSIONAL DEFINITION

A concentric model of financial services participants with AI as the shared enabling core.

KEY DETAILS

- Overlap zones = collaboration and competition
- AI sits at the core enabling layer
- Payments, lending, insurance, wealth mgmt
- Dynamic boundaries shifting continuously

ILLUSTRATIVE FIGURE

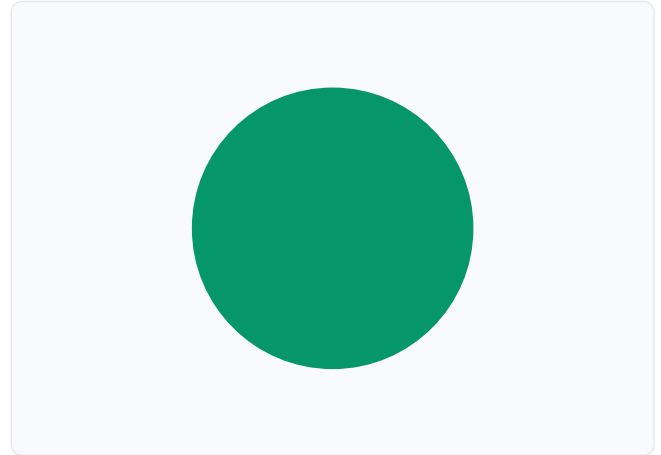


Figure 13.1 — Concentric circles: banks → fintechs → big tech with AI core

SLIBRARY 13

AI in Financial Markets

SOURCE (HARVARD REFERENCING STYLE)

Philippon, T. (2019) 'On fintech and financial inclusion', BIS Working Papers, No. 841. Basel: BIS.

AI Trading Architecture

SLIBRARY 13 · SHEET 2 / 5

Data feeds → signal generation → execution with risk controls

PROFESSIONAL DEFINITION

A layered system design for algorithmic trading: data infrastructure, model layer, and strategy layer with risk controls.

KEY DETAILS

- Horizontal layers: infra, models, strategy
- Latency and risk controls at each layer
- Low-latency vs. fundamental approaches
- López de Prado (2018) framework

ILLUSTRATIVE FIGURE

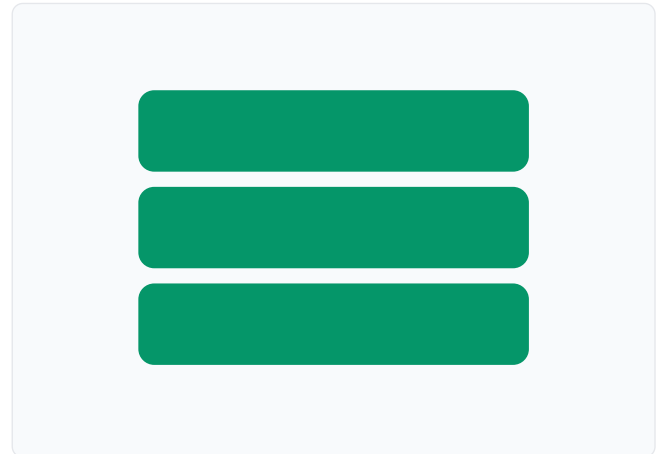


Figure 13.2 — *Data feeds → signal generation → execution with risk controls*

SLIBRARY 13

AI in Financial Markets

SOURCE (HARVARD REFERENCING STYLE)

López de Prado, M. (2018) *Advances in Financial Machine Learning*. Hoboken: Wiley.

Democratisation Ladder

SLIBRARY 13 · SHEET 3 / 5

Institutional-only → retail access → open-source AI for all

PROFESSIONAL DEFINITION

A progression model showing how AI lowers barriers to financial market participation.

KEY DETAILS

- Each rung lowers barriers to entry
- Robo-advisors enabled retail access
- Open-source models empower all investors
- Ongoing regulatory considerations

ILLUSTRATIVE FIGURE

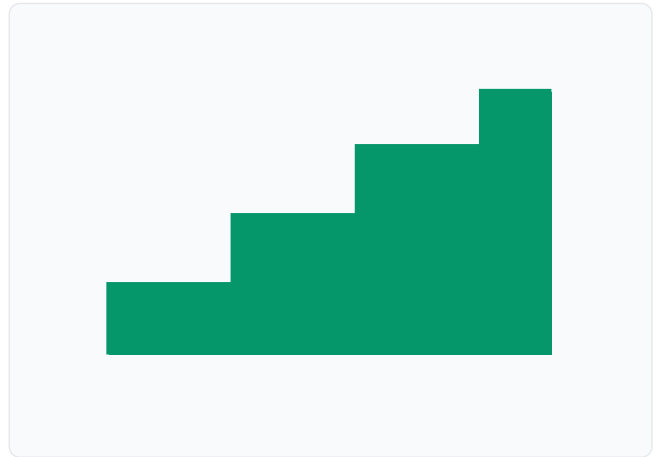


Figure 13.3 — *Institutional-only → retail access → open-source AI for all*

SLIBRARY 13

AI in Financial Markets

SOURCE (HARVARD REFERENCING STYLE)

Vives, X. (2019) 'Digital disruption in banking', Annual Review of Financial Economics, 11, pp. 243–272.

Startup Funding Funnel

SLIBRARY 13 · SHEET 4 / 5

Pre-seed → Seed → Series A/B → IPO with survival rates

PROFESSIONAL DEFINITION

A narrowing pipeline from pre-seed through IPO, showing declining survival rates at each stage.

KEY DETAILS

- Narrowing funnel: survival rates drop
- AI startup-specific failure points
- OpenClaw case study application
- Capital efficiency metrics at each stage

ILLUSTRATIVE FIGURE



Figure 13.4 — *Pre-seed → Seed → Series A/B → IPO with survival rates*

SLIBRARY 13

AI in Financial Markets

SOURCE (HARVARD REFERENCING STYLE)

Gompers, P.A. et al. (2020) 'How do venture capitalists make decisions?', *Journal of Financial Economics*, 135(1), pp. 169–190.

RegTech Compliance Loop

SLIBRARY 13 · SHEET 5 / 5

Monitor → detect → report → adapt cycle automated by AI

PROFESSIONAL DEFINITION

A continuous regulatory cycle—monitor, detect, report, adapt—where AI automates each node.

KEY DETAILS

- Circular: continuous compliance
- AI automates each node in the loop
- Regulatory change triggers adaptation
- Reduces compliance cost 30–50%

ILLUSTRATIVE FIGURE

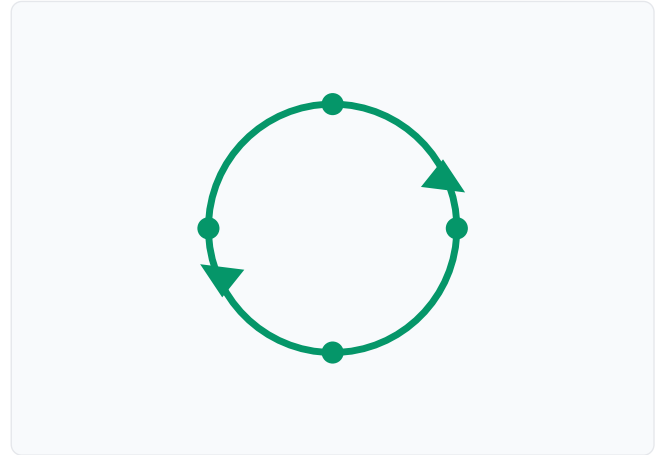


Figure 13.5 — *Monitor → detect → report → adapt cycle automated by AI*

SLIBRARY 13

AI in Financial Markets

SOURCE (HARVARD REFERENCING STYLE)

Arner, D.W., Barberis, J. and Buckley, R.P. (2017) 'FinTech, RegTech, and the reconceptualization of financial regulation', *Northwestern Journal of International Law & Business*, 37(3), pp. 371–413.

Three Lines of Defence

SLIBRARY 14 · SHEET 1 / 5

Business units → risk/compliance → internal audit

PROFESSIONAL DEFINITION

A governance model where risk ownership sits with business units (1st), oversight with compliance (2nd), and assurance with audit (3rd).

KEY DETAILS

- 1st line: business units own risk
- 2nd line: risk and compliance oversight
- 3rd line: internal audit assurance
- AI model governance mapped to each

ILLUSTRATIVE FIGURE

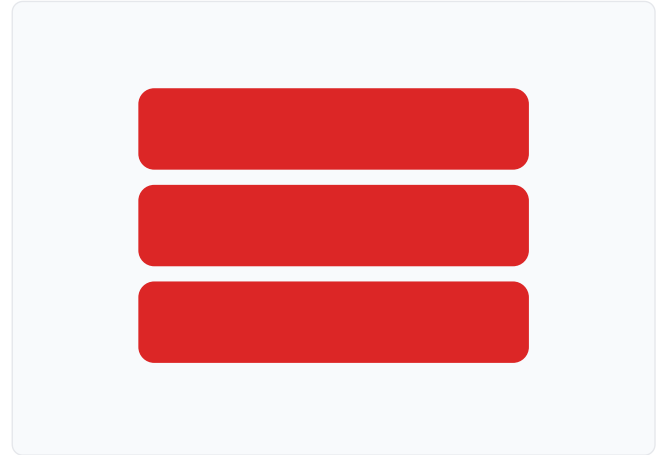


Figure 14.1 — *Business units → risk/compliance → internal audit*

SLIBRARY 14

AI Risk Management

SOURCE (HARVARD REFERENCING STYLE)

Institute of Internal Auditors (2020) The IIA's Three Lines Model. Lake Mary: IIA.

Model Risk Taxonomy

SLIBRARY 14 · SHEET 2 / 5

Data risk, design risk, implementation risk with sub-categories

PROFESSIONAL DEFINITION

A classification tree of AI model risks—data risk, design risk, and implementation risk—with sub-categories.

KEY DETAILS

- Tree structure with three main branches
- Each branch has sub-categories
- Prioritises validation effort
- SR 11-7 / Basel framework aligned

ILLUSTRATIVE FIGURE

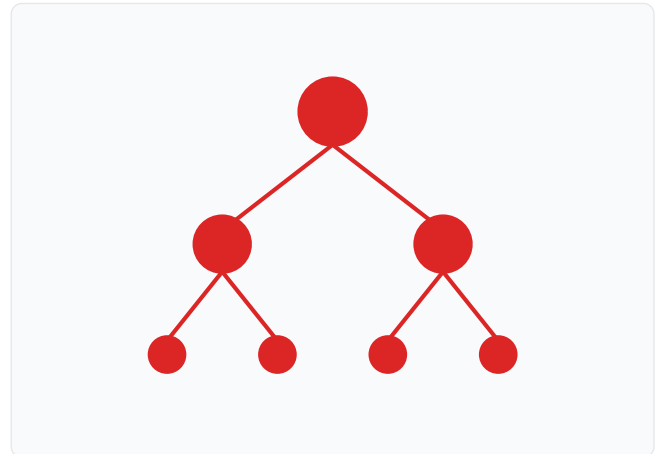


Figure 14.2 — *Data risk, design risk, implementation risk with sub-categories*

SLIBRARY 14

AI Risk Management

SOURCE (HARVARD REFERENCING STYLE)

Board of Governors of the Federal Reserve System (2011) SR 11-7: Guidance on Model Risk Management. Washington, DC.

Stress Testing Framework

SLIBRARY 14 · SHEET 3 / 5

Scenario → Model → Impact → Capital buffer pipeline

PROFESSIONAL DEFINITION

A pipeline for resilience testing: define adverse scenarios, run models, assess impact, determine capital buffers.

KEY DETAILS

- Linear pipeline with feedback loop
- AI generates adverse scenarios faster
- Links to capital adequacy requirements
- Connects to NIST AI RMF (Slibrary 1)

ILLUSTRATIVE FIGURE

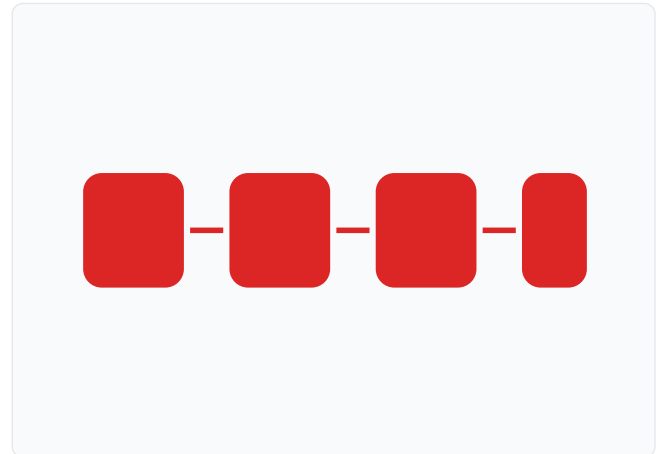


Figure 14.3 — *Scenario → Model → Impact → Capital buffer pipeline*

SLIBRARY 14

AI Risk Management

SOURCE (HARVARD REFERENCING STYLE)

Basel Committee on Banking Supervision (2018) Stress Testing Principles. Basel: BIS.

Algorithmic Bias Audit

SLIBRARY 14 · SHEET 4 / 5

Define → Measure → Mitigate → Monitor continuous cycle

PROFESSIONAL DEFINITION

A continuous cycle—define, measure, mitigate, monitor—to ensure AI fairness.

KEY DETAILS

- Circular continuous improvement
- Protected attributes checklist
- Fairness metrics: disparate impact ratio
- Regulatory requirement in EU AI Act

ILLUSTRATIVE FIGURE

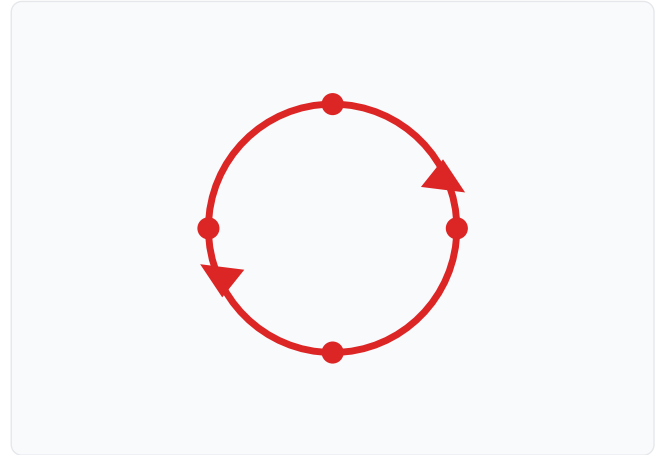


Figure 14.4 — *Define → Measure → Mitigate → Monitor continuous cycle*

SLIBRARY 14

AI Risk Management

SOURCE (HARVARD REFERENCING STYLE)

Barocas, S., Hardt, M. and Narayanan, A. (2023) *Fairness and Machine Learning*. Cambridge: MIT Press.

Black Swan Radar

SLIBRARY 14 · SHEET 5 / 5

Concentric risk rings from known-knowns to unknown-unknowns

PROFESSIONAL DEFINITION

A concentric risk visualisation mapping threats from known-knowns to unknown-unknowns.

KEY DETAILS

- Centre: known-knowns (managed)
- Middle: known-unknowns (hedged)
- Outer: unknown-unknowns (tail risk)
- AI improves detection but can amplify

ILLUSTRATIVE FIGURE

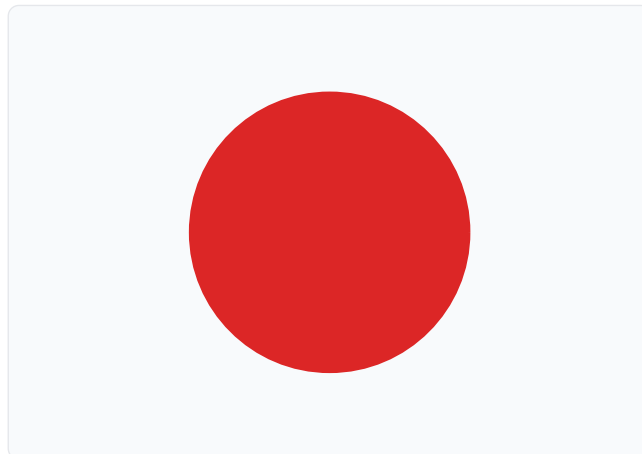


Figure 14.5 — Concentric risk rings from known-knowns to unknown-unknowns

SLIBRARY 14

AI Risk Management

SOURCE (HARVARD REFERENCING STYLE)

Taleb, N.N. (2007) The Black Swan: The Impact of the Highly Improbable. New York: Random House.

AI Talent Gap Heatmap

SLIBRARY 15 · SHEET 1 / 5

Roles × regions colour-coded by shortage severity

PROFESSIONAL DEFINITION

A matrix mapping AI roles against regions, colour-coded by shortage severity.

KEY DETAILS

- Data engineers, ML engineers most scarce
- Finance sector competes with big tech
- Informs hiring vs. upskilling decisions
- Geographic arbitrage opportunities

ILLUSTRATIVE FIGURE

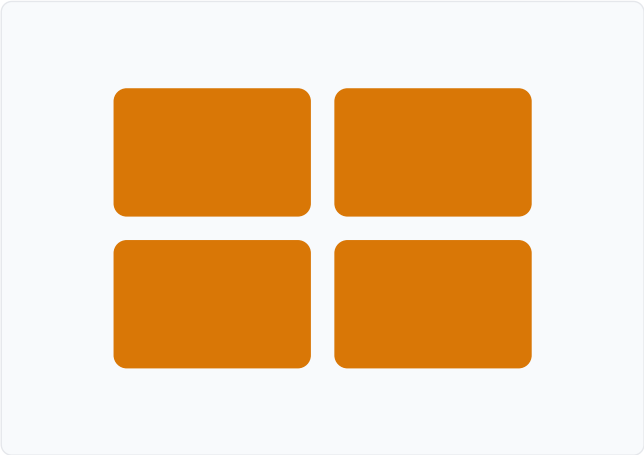


Figure 15.1 — Roles × regions colour-coded by shortage severity

SLIBRARY 15

Talent & Resources

SOURCE (HARVARD REFERENCING STYLE)

World Economic Forum (2023) The Future of Jobs Report 2023. Geneva: WEF.

Build-Buy-Borrow Matrix

SLIBRARY 15 · SHEET 2 / 5

Strategic vs. commodity skills × build/hire/contract options

PROFESSIONAL DEFINITION

A workforce planning matrix crossing skill type with sourcing mode.

KEY DETAILS

- Build (train), buy (hire), borrow (contract)
- Links to AI project lifecycle (Slibrary 3)
- Dynamic rebalancing as needs evolve
- Practical workforce planning tool

ILLUSTRATIVE FIGURE

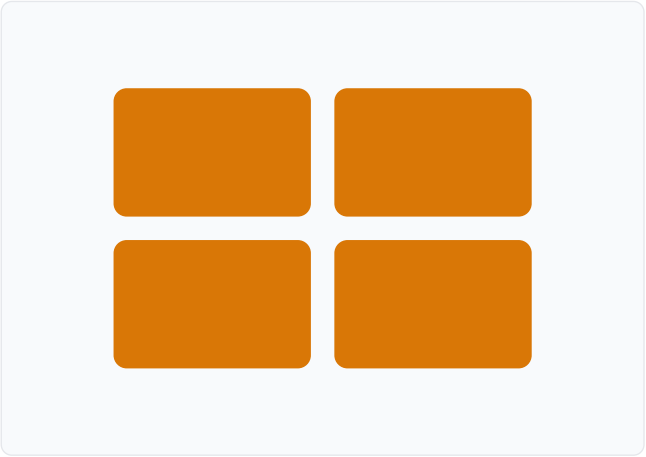


Figure 15.2 — Strategic vs. commodity skills × build/hire/contract options

SLIBRARY 15

Talent & Resources

SOURCE (HARVARD REFERENCING STYLE)

Cappelli, P. (2008) Talent on Demand. Boston: Harvard Business Press.

Skills Adjacency Map

SLIBRARY 15 · SHEET 3 / 5

Network graph of current → AI skills with reskilling paths

PROFESSIONAL DEFINITION

A network graph connecting existing competencies to target AI skills via shortest reskilling paths.

KEY DETAILS

- Short hops = easier reskilling paths
- Identifies bridge roles
- Reduces time-to-competency
- Network graph visualisation

ILLUSTRATIVE FIGURE

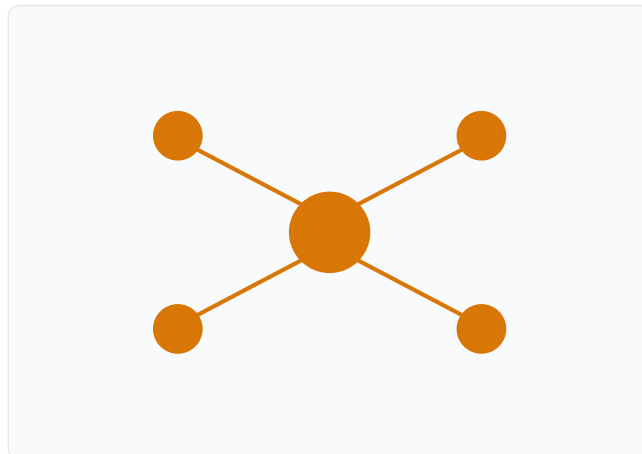


Figure 15.3 — Network graph of current → AI skills with reskilling paths

SLIBRARY 15

Talent & Resources

SOURCE (HARVARD REFERENCING STYLE)

World Economic Forum (2023) The Future of Jobs Report 2023. Geneva: WEF.

Total Cost of AI Ownership

SLIBRARY 15 · SHEET 4 / 5

Iceberg: visible costs above, hidden costs 3–5× below waterline

PROFESSIONAL DEFINITION

An iceberg model revealing that visible AI costs represent only a fraction of true cost, with hidden expenses 3–5× larger.

KEY DETAILS

- Above waterline: compute, licences
- Below: data cleaning, retraining, governance
- True cost is 3–5× initial estimate
- Critical for investment business cases

ILLUSTRATIVE FIGURE

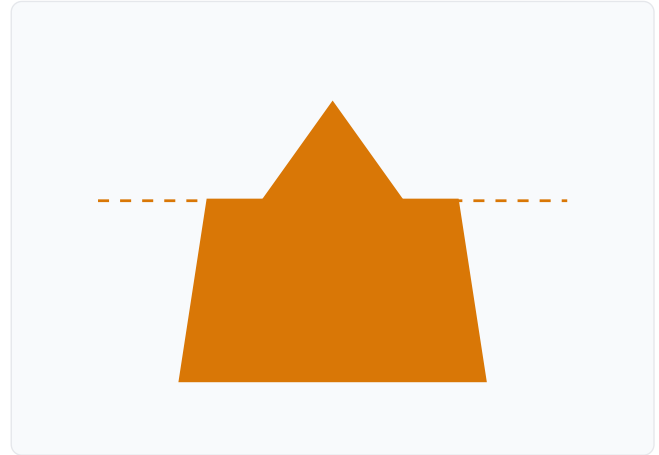


Figure 15.4 — *Iceberg: visible costs above, hidden costs 3–5× below waterline*

SLIBRARY 15

Talent & Resources

SOURCE (HARVARD REFERENCING STYLE)

Iansiti, M. and Lakhani, K.R. (2020) Competing in the Age of AI. Boston: Harvard Business Review Press.

Org Design for AI Teams

SLIBRARY 15 · SHEET 5 / 5

Hub-and-spoke vs. embedded vs. CoE models

PROFESSIONAL DEFINITION

Three organisational archetypes: hub-and-spoke, embedded, and Centre of Excellence.

KEY DETAILS

- Three organisational archetypes
- Trade-offs: speed vs. consistency vs. scale
- Match structure to AI maturity stage
- No single best model for all firms

ILLUSTRATIVE FIGURE

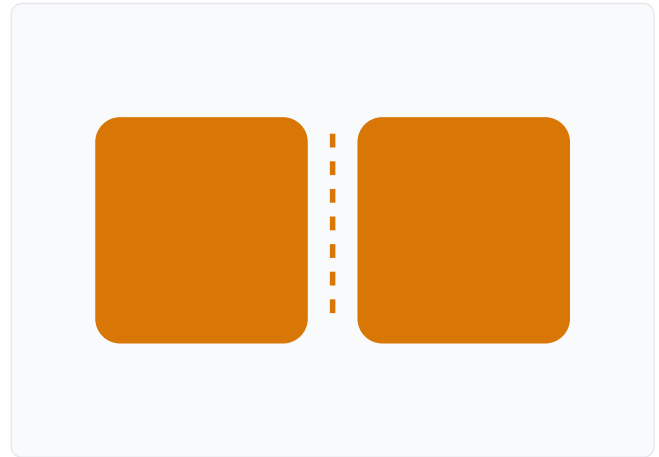


Figure 15.5 — *Hub-and-spoke vs. embedded vs. CoE models*

SLIBRARY 15

Talent & Resources

SOURCE (HARVARD REFERENCING STYLE)

Fountain, T., McCarthy, B. and Saleh, T. (2019) 'Building the AI-powered organization', Harvard Business Review, 97(4), pp. 62–73.

AI Carbon Footprint Stack

SLIBRARY 16 · SHEET 1 / 5

Hardware → data centre → training → inference CO₂ layers

PROFESSIONAL DEFINITION

A layered model quantifying carbon emissions across hardware, data centre, training, and inference.

KEY DETAILS

- Each layer's carbon contribution
- Training GPT-scale ≈ transatlantic flights
- Inference costs accumulate at scale
- Strubell et al. (2019) research

ILLUSTRATIVE FIGURE

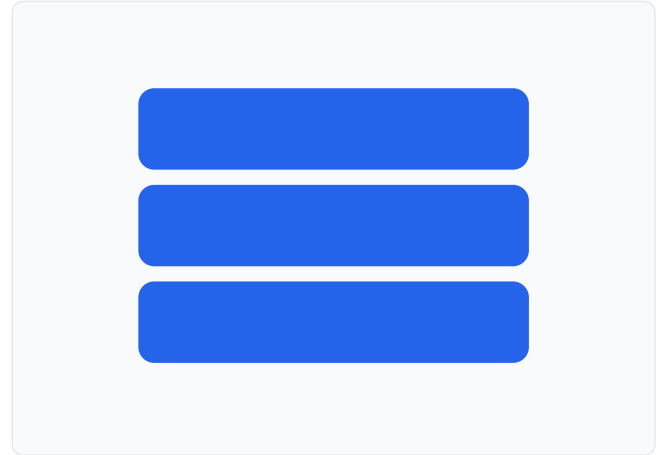


Figure 16.1 — Hardware → data centre → training → inference CO₂ layers

SLIBRARY 16

Sustainable & Green AI

SOURCE (HARVARD REFERENCING STYLE)

Strubell, E., Ganesh, A. and McCallum, A. (2019) 'Energy and policy considerations for deep learning in NLP', Proceedings of the 57th Annual Meeting of the ACL, pp. 3645–3650.

Green AI vs. Red AI

SLIBRARY 16 · SHEET 2 / 5

Red AI (accuracy at any cost) ← → Green AI (efficiency-first)

PROFESSIONAL DEFINITION

A distinction between Red AI (maximising accuracy regardless of cost) and Green AI (prioritising efficiency).

KEY DETAILS

- Red AI maximises accuracy regardless
- Green AI optimises for efficiency
- Trade-off zone in the middle
- Schwartz et al. (2020) distinction

ILLUSTRATIVE FIGURE

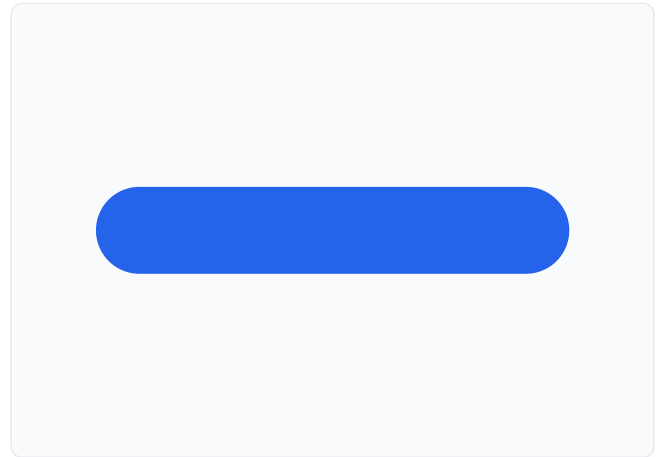


Figure 16.2 — *Red AI (accuracy at any cost) ← → Green AI (efficiency-first)*

SLIBRARY 16

Sustainable & Green AI

SOURCE (HARVARD REFERENCING STYLE)

Schwartz, R. et al. (2020) 'Green AI', Communications of the ACM, 63(12), pp. 54–63.

ESG-AI Integration Matrix

SLIBRARY 16 · SHEET 3 / 5

E, S, G rows × AI opportunity and risk columns

PROFESSIONAL DEFINITION

A grid mapping E, S, G dimensions against AI opportunity and risk columns.

KEY DETAILS

- Where AI supports each ESG dimension
- Where AI creates ESG risk
- Guides responsible deployment
- Connects sustainability to strategy

ILLUSTRATIVE FIGURE

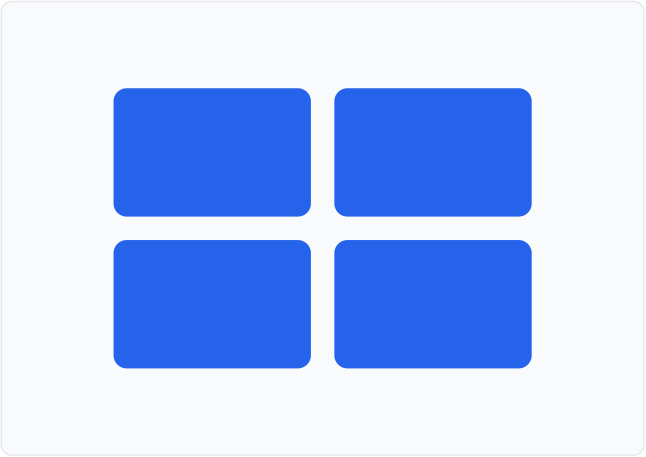


Figure 16.3 — E, S, G rows × AI opportunity and risk columns

SLIBRARY 16

Sustainable & Green AI

SOURCE (HARVARD REFERENCING STYLE)

Vinuesa, R. et al. (2020) 'The role of artificial intelligence in achieving the SDGs', Nature Communications, 11(1), p. 233.

Compute Efficiency Curve

SLIBRARY 16 · SHEET 4 / 5

Model size vs. diminishing performance gains

PROFESSIONAL DEFINITION

A diminishing returns curve showing performance gains flatten as computational cost increases.

KEY DETAILS

- X: model size / FLOPs
- Y: performance (diminishing returns)
- Optimal point before waste zone
- Argues for right-sized models

ILLUSTRATIVE FIGURE

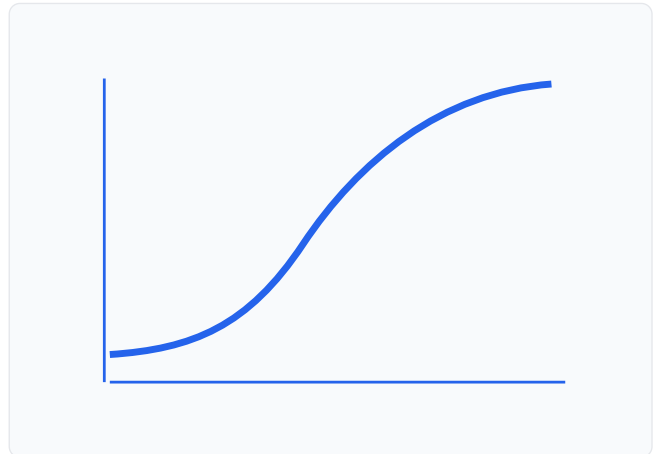


Figure 16.4 — *Model size vs. diminishing performance gains*

SLIBRARY 16

Sustainable & Green AI

SOURCE (HARVARD REFERENCING STYLE)

Kaplan, J. et al. (2020) 'Scaling laws for neural language models', arXiv preprint arXiv:2001.08361.

Responsible AI Scorecard

SLIBRARY 16 · SHEET 5 / 5

Dashboard: fairness, transparency, privacy, safety traffic lights

PROFESSIONAL DEFINITION

A governance dashboard scoring AI systems across fairness, transparency, privacy, and safety using traffic-light indicators.

KEY DETAILS

- Traffic-light scoring per dimension
- Aggregated into org-level rating
- Connects to NIST AI RMF (Slibrary 1)
- Reporting tool for leadership

ILLUSTRATIVE FIGURE

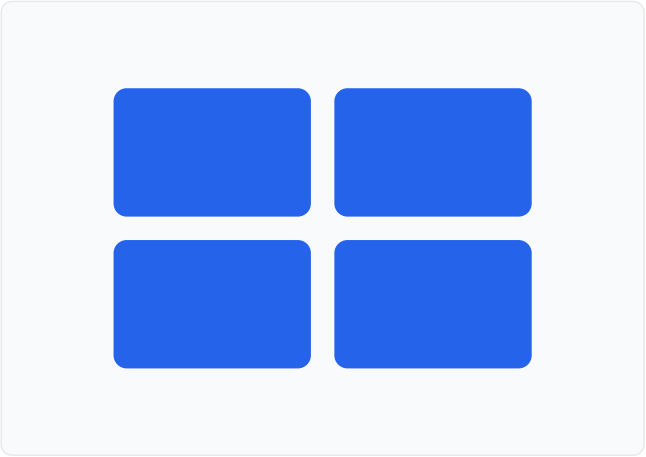


Figure 16.5 — Dashboard: fairness, transparency, privacy, safety traffic lights

SLIBRARY 16

Sustainable & Green AI

SOURCE (HARVARD REFERENCING STYLE)

NIST (2023) AI Risk Management Framework (AI RMF 1.0). NIST AI 100-1.

Authorship Decision Tree

SLIBRARY 17 · SHEET 1 / 5

Is there a human author? Sufficient creative control?

PROFESSIONAL DEFINITION

A branching logic for determining copyrightability of AI-involved works.

KEY DETAILS

- Branching logic for copyrightability
- AI as tool vs. autonomous creator
- US Copyright Office approach
- EU emerging framework

ILLUSTRATIVE FIGURE

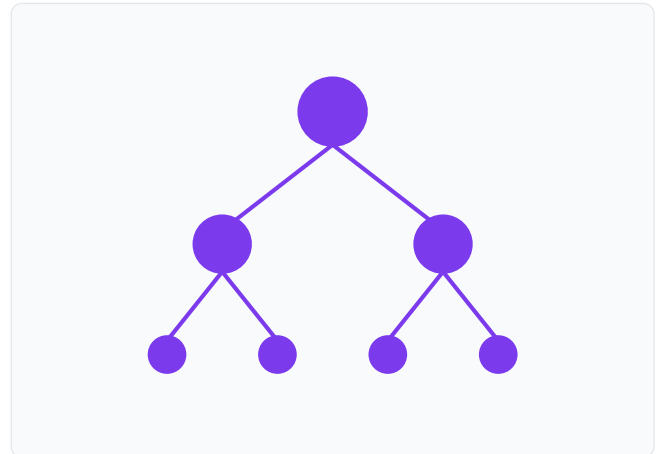


Figure 17.1 — *Is there a human author? Sufficient creative control?*

SLIBRARY 17

AI & Copyright

SOURCE (HARVARD REFERENCING STYLE)

US Copyright Office (2023) Copyright Registration Guidance: Works Containing Material Generated by AI. 88 Fed. Reg. 16190.

AI-Generated Works Spectrum

SLIBRARY 17 · SHEET 2 / 5

Human-created → AI-assisted → AI-directed → autonomous

PROFESSIONAL DEFINITION

A continuum of human involvement in AI-assisted creation, from fully human-created to fully autonomous.

KEY DETAILS

- Protection decreases left to right
- Key boundary: sufficient human control
- Different jurisdictions, different lines
- Rapidly evolving legal landscape

ILLUSTRATIVE FIGURE

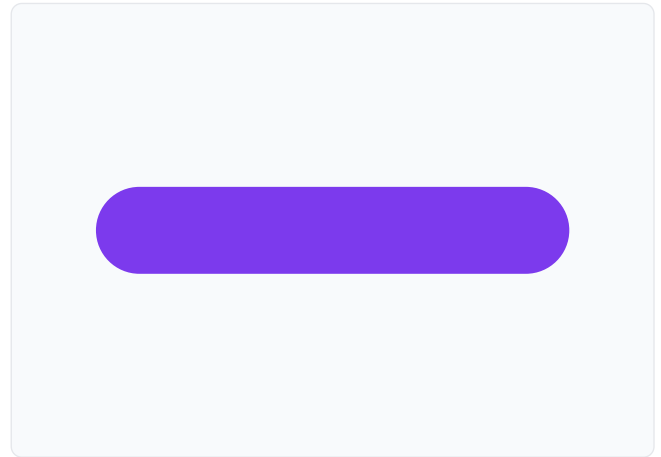


Figure 17.2 — *Human-created → AI-assisted → AI-directed → autonomous*

SLIBRARY 17

AI & Copyright

SOURCE (HARVARD REFERENCING STYLE)

Lee, J.A. et al. (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

TDM Exception Flowchart

SLIBRARY 17 · SHEET 3 / 5

EU Digital Single Market Art.3–4 decision logic

PROFESSIONAL DEFINITION

A decision flowchart for EU text and data mining exceptions under Articles 3 and 4 DSM Directive.

KEY DETAILS

- Art.3: research exception (broad)
- Art.4: commercial use + opt-out check
- Key for AI training data legality
- Rightsholder opt-out mechanisms

ILLUSTRATIVE FIGURE

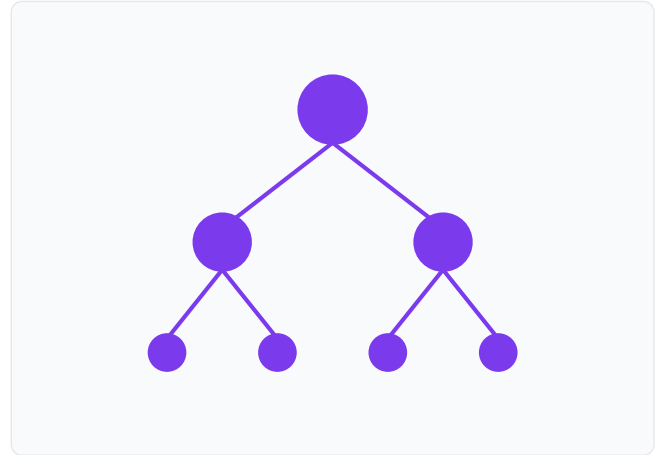


Figure 17.3 — EU Digital Single Market Art.3–4 decision logic

SLIBRARY 17

AI & Copyright

SOURCE (HARVARD REFERENCING STYLE)

European Parliament (2019) Directive (EU) 2019/790 on copyright in the Digital Single Market.

Fair Use Four-Factor Test

SLIBRARY 17 · SHEET 4 / 5

Purpose, nature, amount, and market effect analysis

PROFESSIONAL DEFINITION

The US copyright defence weighing purpose, nature, amount, and market effect.

KEY DETAILS

- US copyright defence framework
- Each factor weighed independently
- No single factor is dispositive
- AI training cases actively litigated

ILLUSTRATIVE FIGURE

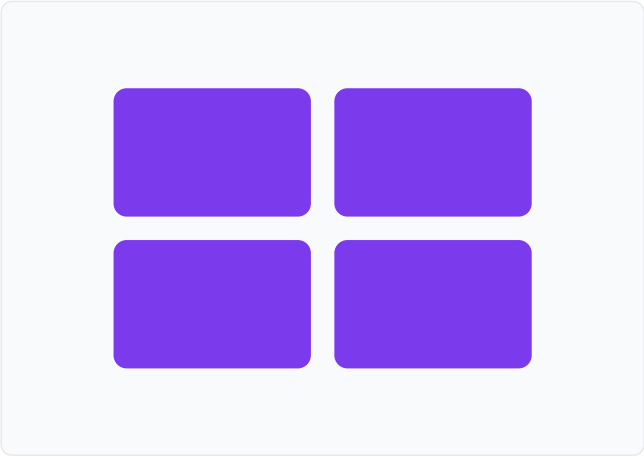


Figure 17.4 — *Purpose, nature, amount, and market effect analysis*

SLIBRARY 17

AI & Copyright

SOURCE (HARVARD REFERENCING STYLE)

17 U.S.C. § 107. See also: Authors Guild v. Google, Inc., 804 F.3d 202 (2d Cir. 2015).

Global Copyright Comparison

SLIBRARY 17 · SHEET 5 / 5

US × EU × UK × China × Japan jurisdiction matrix

PROFESSIONAL DEFINITION

A jurisdiction matrix mapping how major countries treat AI authorship, training data, and output.

KEY DETAILS

- Rows: authorship, training, output rights
- Traffic-light status per jurisdiction
- Snapshot in time — rapidly evolving
- Essential for multinational AI strategy

ILLUSTRATIVE FIGURE

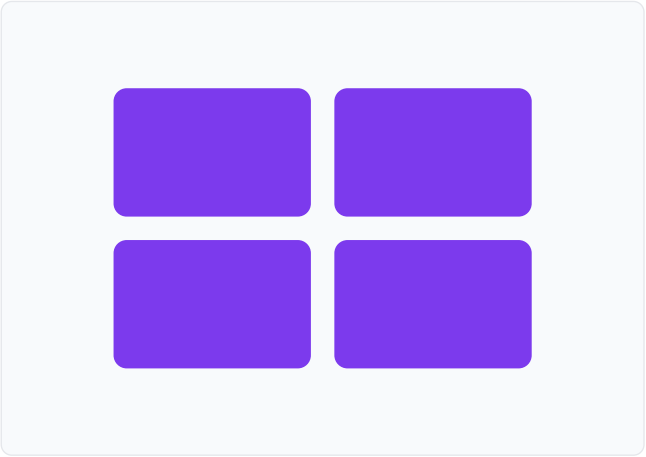


Figure 17.5 — US × EU × UK × China × Japan jurisdiction matrix

SLIBRARY 17

AI & Copyright

SOURCE (HARVARD REFERENCING STYLE)

WIPO (2020) Revised Issues Paper on IP Policy and AI. Geneva: WIPO.

Data IP Protection Layers

SLIBRARY 18 · SHEET 1 / 5

Concentric rings: copyright → database rights → trade secrets → contract

PROFESSIONAL DEFINITION

A concentric model showing how multiple IP regimes can stack to protect the same dataset.

KEY DETAILS

- Multiple protection layers can stack
- No single regime covers all data
- Technical protection as outer layer
- Layered strategy for maximum coverage

ILLUSTRATIVE FIGURE

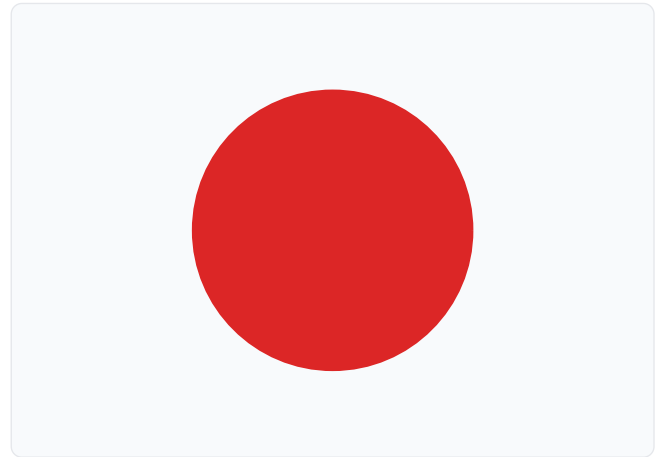


Figure 18.1 — *Concentric rings: copyright → database rights → trade secrets → contract*

SLIBRARY 18

Data as IP

SOURCE (HARVARD REFERENCING STYLE)

Drexl, J. et al. (2017) 'Data ownership and access to data', Max Planck Institute Research Paper No. 16-10.

Sui Generis Database Rights

SLIBRARY 18 · SHEET 2 / 5

EU substantial investment test with 15-year renewable protection

PROFESSIONAL DEFINITION

An EU-specific IP right protecting databases where there has been substantial investment.

KEY DETAILS

- Extraction vs. re-utilisation rights
- Substantial investment threshold
- 15-year renewable protection
- EU-specific — no US equivalent

ILLUSTRATIVE FIGURE

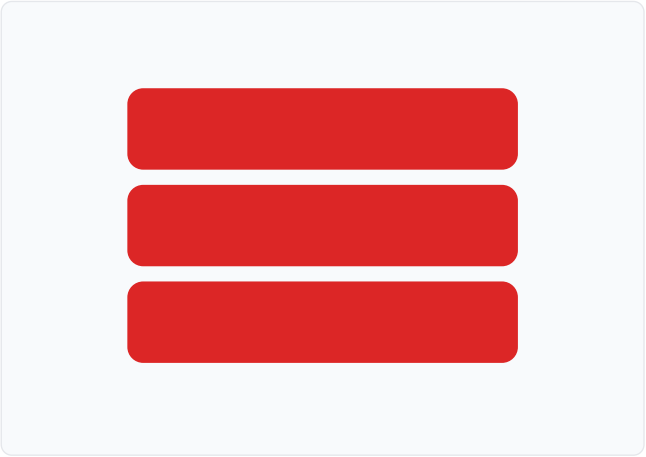


Figure 18.2 — EU substantial investment test with 15-year renewable protection

SLIBRARY 18

Data as IP

SOURCE (HARVARD REFERENCING STYLE)

European Parliament (1996) Directive 96/9/EC on the legal protection of databases.

Trade Secret vs. Patent 2×2

SLIBRARY 18 · SHEET 3 / 5

Disclosure willingness × duration need decision matrix

PROFESSIONAL DEFINITION

A decision matrix comparing two IP protection strategies along disclosure and duration axes.

KEY DETAILS

- Trade secret: no disclosure, indefinite
- Patent: full disclosure, 20 years
- AI models often better as trade secrets
- Strategic choice for each asset

ILLUSTRATIVE FIGURE

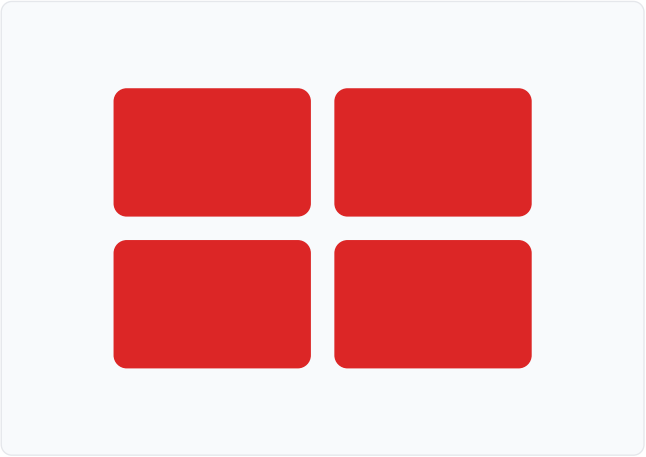


Figure 18.3 — *Disclosure willingness × duration need decision matrix*

SLIBRARY 18 Data as IP

SOURCE (HARVARD REFERENCING STYLE)

Lee, J.A. et al. (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

Data Licensing Model Map

SLIBRARY 18 · SHEET 4 / 5

Open → CC BY → CC BY-SA → proprietary spectrum

PROFESSIONAL DEFINITION

A spectrum of data licensing models from fully open through Creative Commons to fully proprietary.

KEY DETAILS

- Restrictions increase left to right
- Commercial use flags at each level
- Critical for AI training data sourcing
- License compliance automation

ILLUSTRATIVE FIGURE

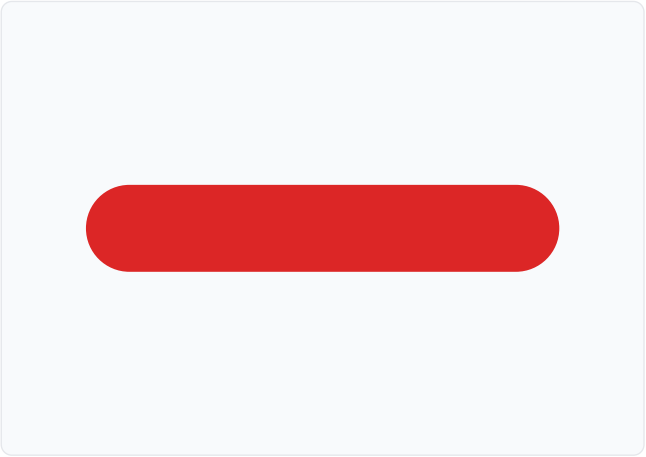


Figure 18.4 — *Open → CC BY → CC BY-SA → proprietary spectrum*

SLIBRARY 18

Data as IP

SOURCE (HARVARD REFERENCING STYLE)

Creative Commons (2023) About CC Licenses. Available at: <https://creativecommons.org/licenses/>.

Open vs. Proprietary Spectrum

SLIBRARY 18 · SHEET 5 / 5

Public domain → freemium → API-gated → exclusive

PROFESSIONAL DEFINITION

A positioning model showing how data sources range from public domain to fully exclusive.

KEY DETAILS

- Competitive advantage increases rightward
- Open data enables but doesn't differentiate
- Proprietary data as modern moat
- Pricing and access models at each level

ILLUSTRATIVE FIGURE

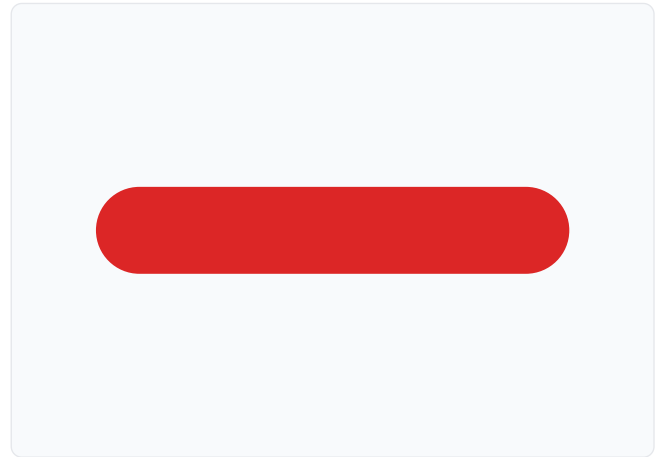


Figure 18.5 — *Public domain → freemium → API-gated → exclusive*

SLIBRARY 18

Data as IP

SOURCE (HARVARD REFERENCING STYLE)

Zuboff, S. (2019) *The Age of Surveillance Capitalism*. London: Profile Books.

Freedom-to-Operate Analysis

SLIBRARY 19 · SHEET 1 / 5

All patents → relevant → blocking → risk assessment funnel

PROFESSIONAL DEFINITION

A filtering process narrowing the patent universe to blocking patents, yielding a risk-rated FTO opinion.

KEY DETAILS

- Narrowing funnel at each filter stage
- Output: FTO opinion (green/amber/red)
- Essential before AI product launch
- AI can automate patent search step

ILLUSTRATIVE FIGURE



Figure 19.1 — *All patents → relevant → blocking → risk assessment funnel*

SLIBRARY 19

IP Strategy & Competition

SOURCE (HARVARD REFERENCING STYLE)

WIPO (2019) Technology Trends 2019: Artificial Intelligence. Geneva: WIPO.

Patent Landscape Map

SLIBRARY 19 · SHEET 2 / 5

Bubble chart: assignees × tech domains with white spaces

PROFESSIONAL DEFINITION

A bubble chart plotting patent assignees against technology domains, revealing competitive white spaces.

KEY DETAILS

- Bubble size = patent portfolio size
- White spaces = opportunity zones
- AI automates landscape generation
- Strategic planning visualisation

ILLUSTRATIVE FIGURE

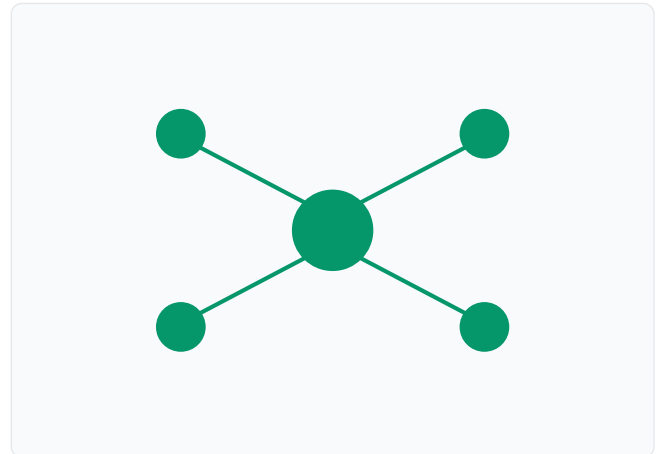


Figure 19.2 — *Bubble chart: assignees × tech domains with white spaces*

SLIBRARY 19

IP Strategy & Competition

SOURCE (HARVARD REFERENCING STYLE)

WIPO (2019) Technology Trends 2019: Artificial Intelligence. Geneva: WIPO.

IP Portfolio Valuation

SLIBRARY 19 · SHEET 3 / 5

Cost → market → income approach comparison

PROFESSIONAL DEFINITION

Three approaches to valuing IP: cost, market, and income.

KEY DETAILS

- Three valuation methodologies
- AI patents: short lifecycle challenge
- Income approach most common for tech
- Links to Unit 6 investment valuation

ILLUSTRATIVE FIGURE

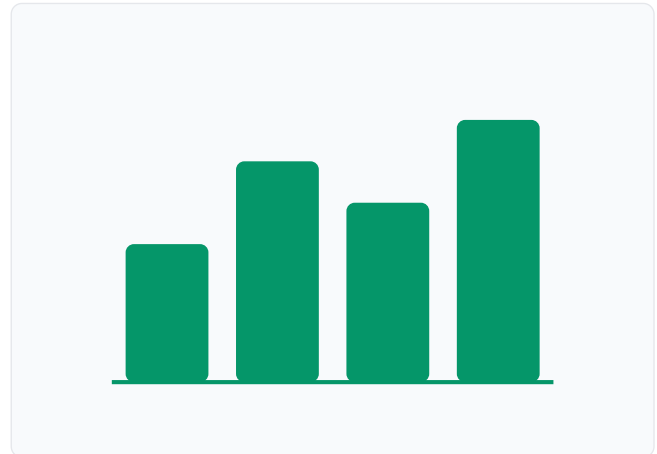


Figure 19.3 — *Cost → market → income approach comparison*

SLIBRARY 19

IP Strategy & Competition

SOURCE (HARVARD REFERENCING STYLE)

Smith, G.V. and Parr, R.L. (2005) Intellectual Property: Valuation, Exploitation, and Infringement Damages. 4th edn. Hoboken: Wiley.

Competition-IP Tension Model

SLIBRARY 19 · SHEET 4 / 5

IP monopoly ← → free competition pulling apart

PROFESSIONAL DEFINITION

A model showing the inherent tension between IP monopoly rights and competition policy.

KEY DETAILS

- Two opposing forces in tension
- Balance shifts with regulation
- EU AI Act adds new constraints
- GPAI obligations layer

ILLUSTRATIVE FIGURE

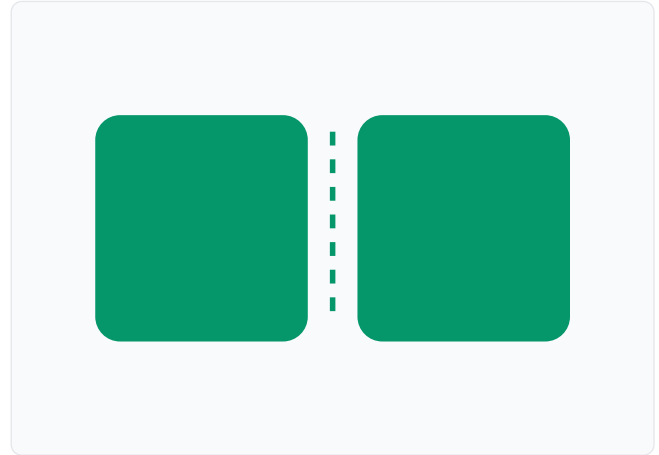


Figure 19.4 — *IP monopoly ← → free competition pulling apart*

SLIBRARY 19

IP Strategy & Competition

SOURCE (HARVARD REFERENCING STYLE)

Shapiro, C. (2001) 'Navigating the patent thicket', in Innovation Policy and the Economy. Cambridge: MIT Press.

FRAND Licensing Diagram

SLIBRARY 19 · SHEET 5 / 5

Fair, Reasonable, Non-Discriminatory for standard-essential patents

PROFESSIONAL DEFINITION

A negotiation framework for licensing standard-essential patents on Fair, Reasonable, and Non-Discriminatory terms.

KEY DETAILS

- Licensors ↔ licensee negotiation flow
- Standard-essential patent context
- Relevant for AI interoperability
- Potential AI standard patents emerging

ILLUSTRATIVE FIGURE

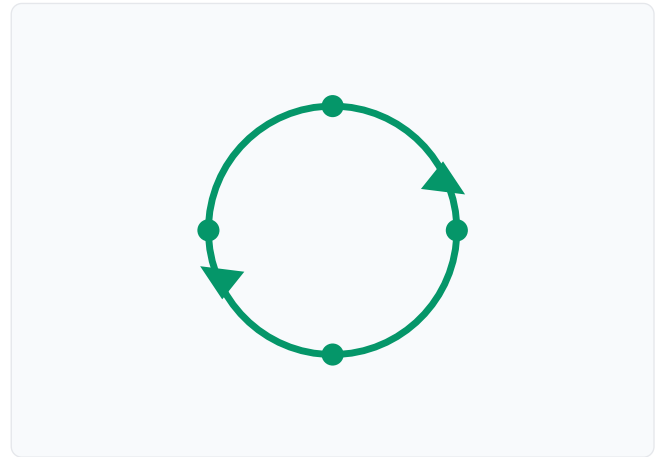


Figure 19.5 — *Fair, Reasonable, Non-Discriminatory for standard-essential patents*

SLIBRARY 19

IP Strategy & Competition

SOURCE (HARVARD REFERENCING STYLE)

Lemley, M.A. (2002) 'Intellectual property rights and standard-setting organizations', *California Law Review*, 90(6), pp. 1889–1980.

AI Liability Chain

SLIBRARY 20 · SHEET 1 / 5

Developer → deployer → user → affected party with liability at each

PROFESSIONAL DEFINITION

A horizontal chain mapping liability from developer through deployer to user and affected party.

KEY DETAILS

- EU AI Liability Directive allocation
- Burden of proof shifts at high-risk tier
- Each node bears specific obligations
- Horizontal chain diagram

ILLUSTRATIVE FIGURE

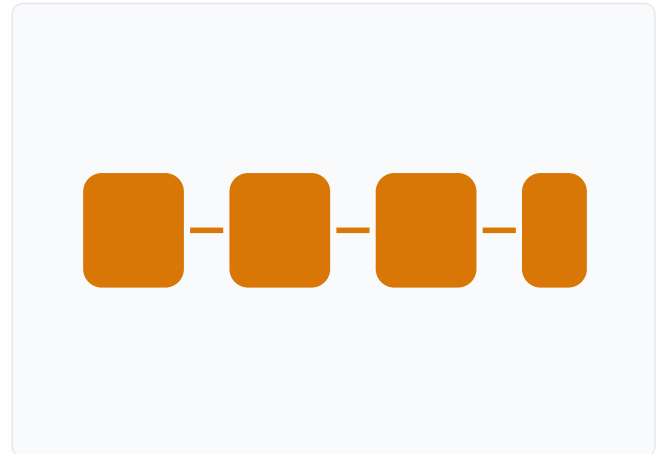


Figure 20.1 — *Developer → deployer → user → affected party with liability at each*

SLIBRARY 20

Liability & Protection

SOURCE (HARVARD REFERENCING STYLE)

European Commission (2022) Proposal for AI Liability Directive. COM(2022) 496.

Product vs. IP Liability 2×2

SLIBRARY 20 · SHEET 2 / 5

Tangible × intangible harm—AI can trigger both

PROFESSIONAL DEFINITION

A matrix distinguishing tangible harm (product liability) from intangible harm (IP liability).

KEY DETAILS

- Product liability: physical damage
- IP liability: infringement, misappropriation
- AI outputs can trigger both simultaneously
- Dual exposure risk for deployers

ILLUSTRATIVE FIGURE

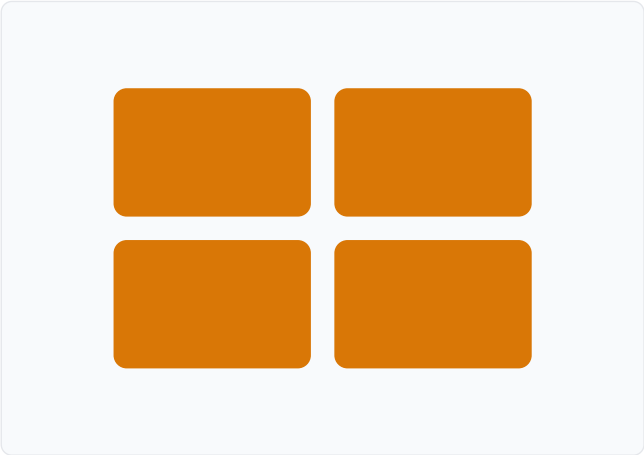


Figure 20.2 — Tangible × intangible harm—AI can trigger both

SLIBRARY 20

Liability & Protection

SOURCE (HARVARD REFERENCING STYLE)

Buiten, M.C. (2019) 'Towards intelligent regulation of AI', European Journal of Risk Regulation, 10(1), pp. 41–59.

Insurance & Indemnity Framework

SLIBRARY 20 · SHEET 3 / 5

Self-insure → contractual indemnity → insurance waterfall

PROFESSIONAL DEFINITION

A risk transfer waterfall from self-insurance through contractual indemnification to third-party insurance.

KEY DETAILS

- Risk transfer waterfall diagram
- AI-specific policies emerging
- Underwriting challenges for novel risks
- Contractual allocation essential

ILLUSTRATIVE FIGURE

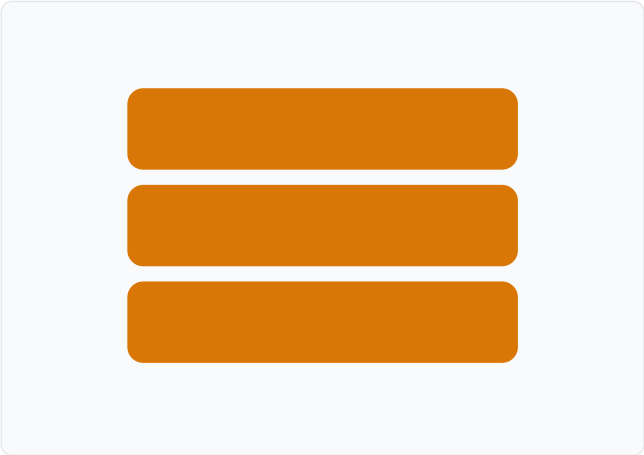


Figure 20.3 — Self-insure → contractual indemnity → insurance waterfall

SLIBRARY 20

Liability & Protection

SOURCE (HARVARD REFERENCING STYLE)

OECD (2021) Recommendation of the Council on Artificial Intelligence. Paris: OECD.

Contractual IP Allocation

SLIBRARY 20 · SHEET 4 / 5

Parties × IP types (foreground, background, sideground)

PROFESSIONAL DEFINITION

A matrix defining ownership of foreground, background, and sideground IP in AI collaborations.

KEY DETAILS

- Who owns what in AI collaborations?
- Joint development vs. work-for-hire
- Template for partnership agreements
- Critical for AI joint ventures

ILLUSTRATIVE FIGURE

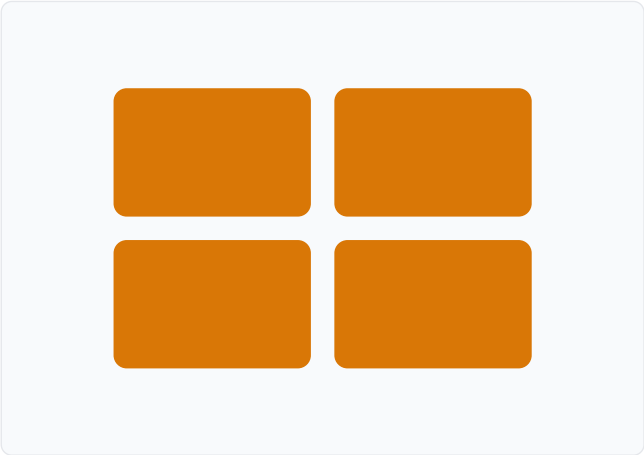


Figure 20.4 — Parties × IP types (foreground, background, sideground)

SLIBRARY 20

Liability & Protection

SOURCE (HARVARD REFERENCING STYLE)

Lee, J.A. et al. (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

Enforcement Decision Flowchart

SLIBRARY 20 · SHEET 5 / 5

Detect → assess strength → cease & desist / negotiate / litigate

PROFESSIONAL DEFINITION

A decision tree: detect infringement, assess strength, then choose enforcement strategy.

KEY DETAILS

- Cost-benefit at each decision node
- AI tools accelerate detection
- Strategic choice: enforce vs. license
- Portfolio-level enforcement strategy

ILLUSTRATIVE FIGURE

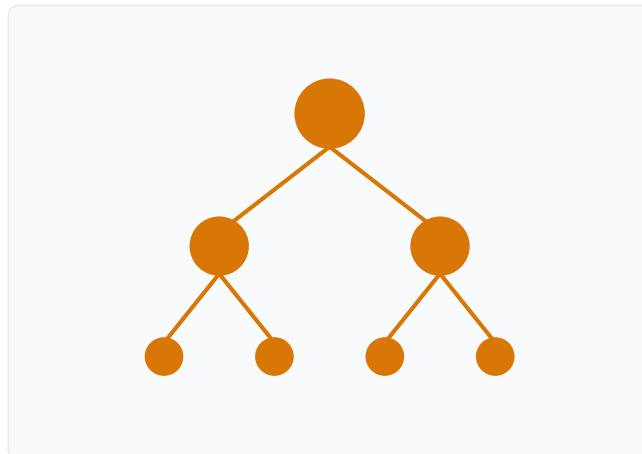


Figure 20.5 — *Detect → assess strength → cease & desist / negotiate / litigate*

SLIBRARY 20

Liability & Protection

SOURCE (HARVARD REFERENCING STYLE)

Lee, J.A. et al. (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

Investment Mgmt Value Chain

SLIBRARY 21 · SHEET 1 / 5

Research → Construction → Execution → Risk → Reporting

PROFESSIONAL DEFINITION

An end-to-end process model for investment management with AI at each node.

KEY DETAILS

- End-to-end investment process
- AI touchpoints at each stage
- Automation potential assessment
- Where human judgment stays essential

ILLUSTRATIVE FIGURE

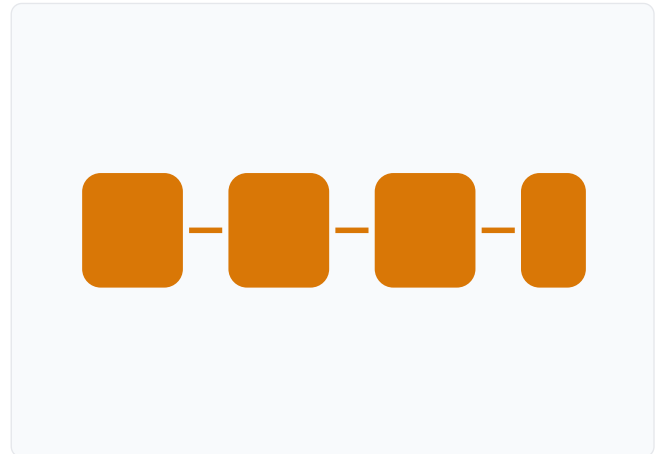


Figure 21.1 — *Research → Construction → Execution → Risk → Reporting*

SLIBRARY 21

AI in Investment Management

SOURCE (HARVARD REFERENCING STYLE)

Muldowney, P. (2023) 'The impact of technology and AI on investment management', *Plans & Trusts*, 41(6), pp. 8–13.

AI Use Case Taxonomy (Finance)

SLIBRARY 21 · SHEET 2 / 5

Front / middle / back office tree with specific applications

PROFESSIONAL DEFINITION

A hierarchical classification of AI applications in finance by office function.

KEY DETAILS

- Tree: office function → use cases
- Leaf nodes: specific AI applications
- Prioritisation heatmap overlay
- Comprehensive scope mapping

ILLUSTRATIVE FIGURE

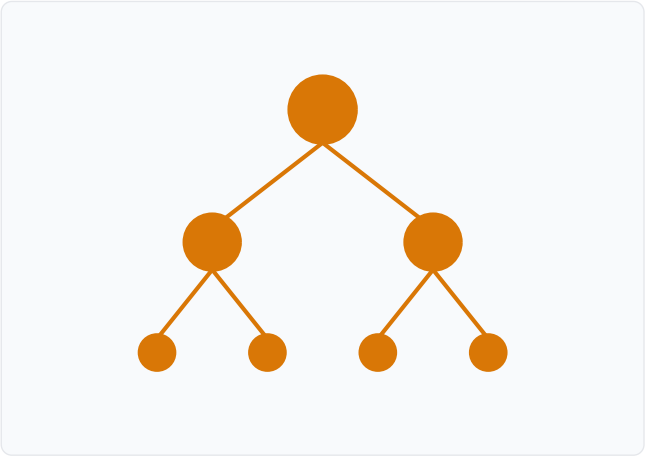


Figure 21.2 — Front / middle / back office tree with specific applications

SLIBRARY 21

AI in Investment Management

SOURCE (HARVARD REFERENCING STYLE)

Cao, L.J. (2022) 'AI in Finance', ACM Computing Surveys, 55(3), pp. 1–38.

Robo-Advisory Architecture

SLIBRARY 21 · SHEET 3 / 5

Client profiling → asset allocation → rebalancing loop

PROFESSIONAL DEFINITION

An automated wealth management system: client profiling, asset allocation, periodic rebalancing.

KEY DETAILS

- Automated feedback loop
- Human override layer on top
- Democratises wealth management
- Regulatory requirements layer

ILLUSTRATIVE FIGURE

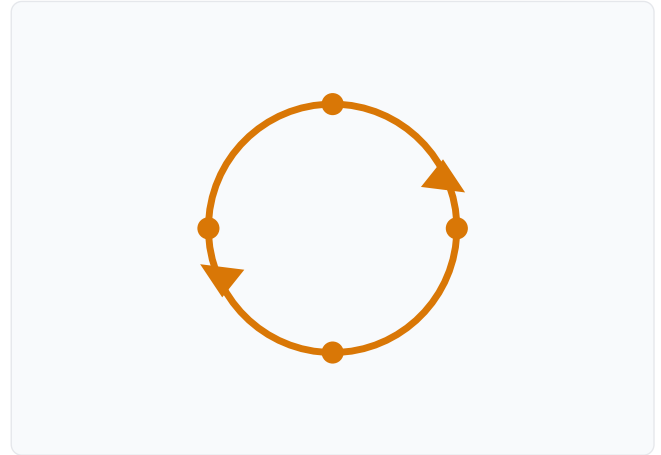


Figure 21.3 — *Client profiling → asset allocation → rebalancing loop*

SLIBRARY 21

AI in Investment Management

SOURCE (HARVARD REFERENCING STYLE)

D'Acunto, F. and Rossi, A.G. (2021) 'New frontiers of robo-advising', Handbook of Financial Technology.

Alpha Generation Pipeline

SLIBRARY 21 · SHEET 4 / 5

Data → signal extraction → portfolio with ML at signal stage

PROFESSIONAL DEFINITION

A systematic investment pipeline where alternative data feeds differentiate returns via ML signal extraction.

KEY DETAILS

- Alternative data feeds as differentiator
- ML at signal extraction stage
- López de Prado (2018) framework
- Systematic vs. discretionary blend

ILLUSTRATIVE FIGURE



Figure 21.4 — *Data → signal extraction → portfolio with ML at signal stage*

SLIBRARY 21

AI in Investment Management

SOURCE (HARVARD REFERENCING STYLE)

López de Prado, M. (2018) *Advances in Financial Machine Learning*. Hoboken: Wiley.

Muldowney ML Adoption Model

SLIBRARY 21 · SHEET 5 / 5

Data cleaning → pattern capture → NLP themes

PROFESSIONAL DEFINITION

A three-stage model of ML adoption in investment: data cleaning, pattern capture, NLP themes—human judgment essential.

KEY DETAILS

- Early: data cleaning and narrowing
- Middle: ML non-linear pattern capture
- Advanced: NLP for emerging themes
- Human judgment remains essential

ILLUSTRATIVE FIGURE

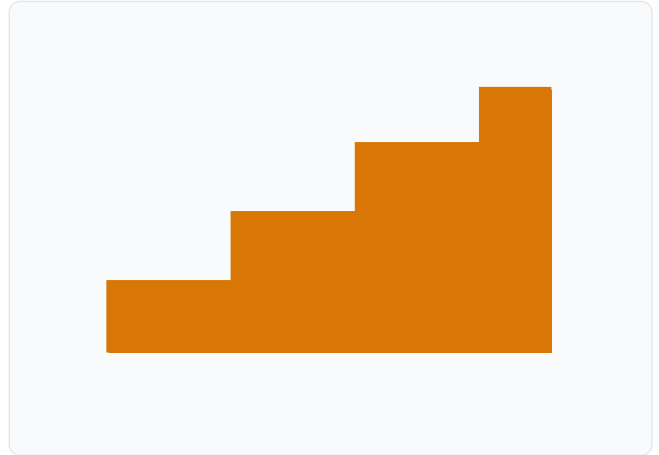


Figure 21.5 — *Data cleaning → pattern capture → NLP themes*

SLIBRARY 21

AI in Investment Management

SOURCE (HARVARD REFERENCING STYLE)

Muldowney, P. (2023) 'The impact of technology and AI on investment management', *Plans & Trusts*, 41(6), pp. 8–13.

MPT Efficient Frontier

SLIBRARY 22 · SHEET 1 / 5

Risk × return curve with capital market line

PROFESSIONAL DEFINITION

A curve representing the set of optimal portfolios offering highest expected return for each risk level.

KEY DETAILS

- X: risk (std dev), Y: expected return
- Curved frontier of optimal portfolios
- Capital Market Line from risk-free rate
- Markowitz (1952); foundation thumbnail

ILLUSTRATIVE FIGURE

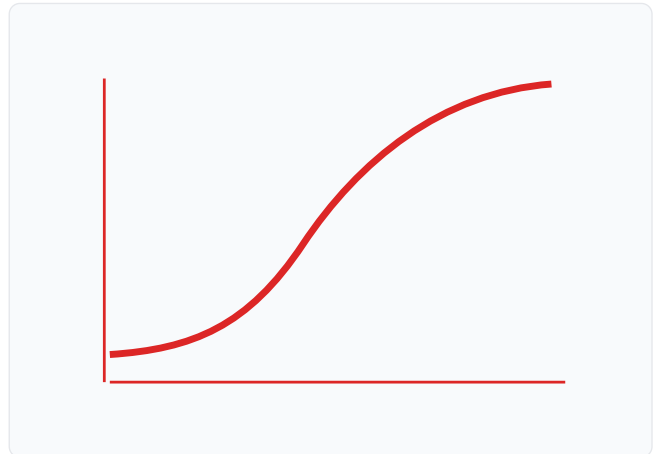


Figure 22.1 — Risk × return curve with capital market line

SLIBRARY 22

Portfolio Optimisation & ML

SOURCE (HARVARD REFERENCING STYLE)

Markowitz, H.M. (1952) 'Portfolio selection', The Journal of Finance, 7(1), pp. 77–91.

Black-Litterman Model

SLIBRARY 22 · SHEET 2 / 5

Market equilibrium + investor views blended into allocation

PROFESSIONAL DEFINITION

A portfolio allocation model blending market equilibrium returns with investor-specific views.

KEY DETAILS

- Two inputs merge into output allocation
- Solves MVO concentration problem
- AI can generate the 'views' input
- Practical portfolio construction tool

ILLUSTRATIVE FIGURE

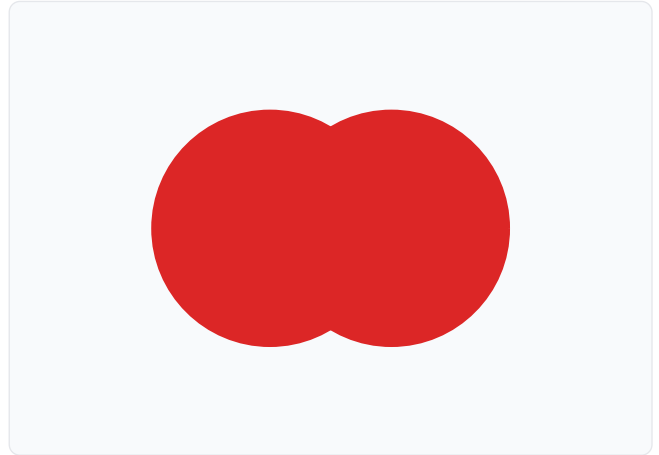


Figure 22.2 — *Market equilibrium + investor views blended into allocation*

SLIBRARY 22

Portfolio Optimisation & ML

SOURCE (HARVARD REFERENCING STYLE)

Black, F. and Litterman, R. (1992) 'Global portfolio optimization', *Financial Analysts Journal*, 48(5), pp. 28–43.

ML-Enhanced Risk-Return 2×2

SLIBRARY 22 · SHEET 3 / 5

Traditional vs. ML-enhanced across four domains

PROFESSIONAL DEFINITION

A matrix comparing traditional and ML-enhanced approaches across return/risk estimation and construction/rebalancing.

KEY DETAILS

- Four quadrants of ML enhancement
- Return estimation, risk estimation
- Portfolio construction, rebalancing
- Shows where ML improves each area

ILLUSTRATIVE FIGURE

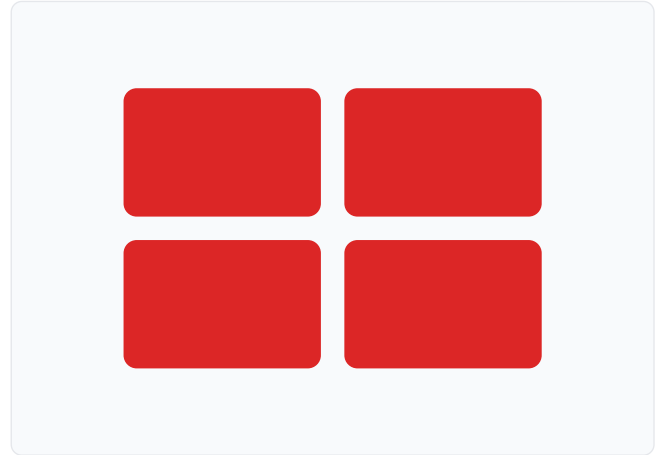


Figure 22.3 — *Traditional vs. ML-enhanced across four domains*

SLIBRARY 22

Portfolio Optimisation & ML

SOURCE (HARVARD REFERENCING STYLE)

Gu, S., Kelly, B. and Xiu, D. (2020) 'Empirical asset pricing via machine learning', *The Review of Financial Studies*, 33(5), pp. 2223–2273.

Factor Investing Architecture

SLIBRARY 22 · SHEET 4 / 5

Value, momentum, quality, size, volatility factor bars

PROFESSIONAL DEFINITION

A framework targeting persistent risk premia with ML enabling dynamic factor timing.

KEY DETAILS

- ML identifies dynamic factor timing
- Smart beta to pure alpha spectrum
- Factor rotation strategies
- Integration with traditional approaches

ILLUSTRATIVE FIGURE

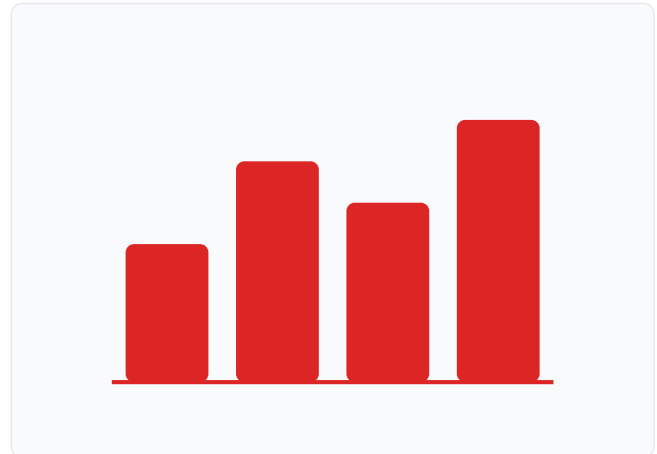


Figure 22.4 — Value, momentum, quality, size, volatility factor bars

SLIBRARY 22

Portfolio Optimisation & ML

SOURCE (HARVARD REFERENCING STYLE)

Ang, A. (2014) Asset Management: A Systematic Approach to Factor Investing. New York: OUP.

Backtesting Validation Loop

SLIBRARY 22 · SHEET 5 / 5

Train → validate → test → deploy → monitor

PROFESSIONAL DEFINITION

A circular validation process with overfitting detection and walk-forward methodology.

KEY DETAILS

- Circular with feedback to retrain
- Overfitting detection at validation
- Walk-forward vs. static split
- Essential for model credibility

ILLUSTRATIVE FIGURE

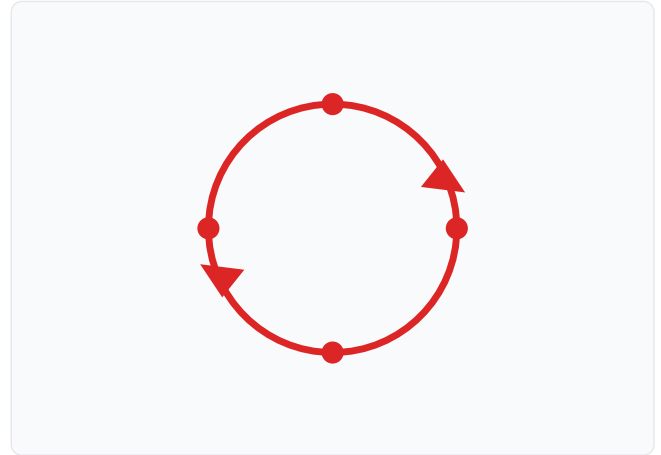


Figure 22.5 — *Train → validate → test → deploy → monitor*

SLIBRARY 22

Portfolio Optimisation & ML

SOURCE (HARVARD REFERENCING STYLE)

López de Prado, M. (2018) *Advances in Financial Machine Learning*. Hoboken: Wiley.

AI Financial Analysis Stack

SLIBRARY 23 · SHEET 1 / 5

Data → features → models → predictions → decisions → execution

PROFESSIONAL DEFINITION

A vertical architecture: data, feature engineering, models, predictions, decisions, execution.

KEY DETAILS

- Vertical stack with human checkpoints
- Each layer builds on the one below
- Feature engineering is the art
- Execution layer closes the loop

ILLUSTRATIVE FIGURE

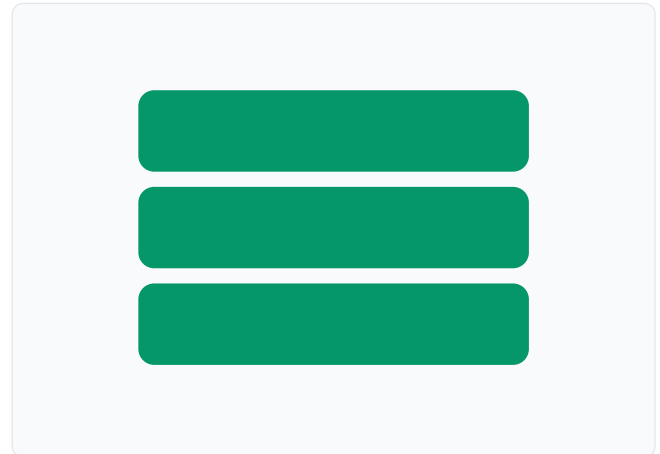


Figure 23.1 — *Data → features → models → predictions → decisions → execution*

SLIBRARY 23

Financial Analysis & Modelling

SOURCE (HARVARD REFERENCING STYLE)

Dixon, M.F., Halperin, I. and Bilokon, P. (2020) Machine Learning in Finance. Cham: Springer.

Sentiment-Driven Alpha

SLIBRARY 23 · SHEET 2 / 5

News/social → NLP → sentiment score → trading signal

PROFESSIONAL DEFINITION

A signal pipeline: news/social → NLP → sentiment scores → trading signals.

KEY DETAILS

- Contrarian vs. momentum strategies
- Real-time vs. batch processing trade-off
- Measurable alpha contribution
- Growing practitioner adoption

ILLUSTRATIVE FIGURE

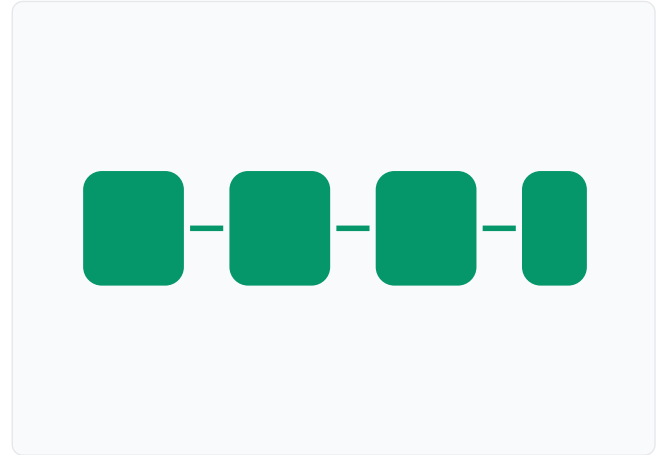


Figure 23.2 — *News/social → NLP → sentiment score → trading signal*

SLIBRARY 23

Financial Analysis & Modelling

SOURCE (HARVARD REFERENCING STYLE)

Loughran, T. and McDonald, B. (2016) 'Textual analysis in accounting and finance', *Journal of Accounting Research*, 54(4), pp. 1187–1230.

Alternative Data Integration

SLIBRARY 23 · SHEET 3 / 5

Satellite, social, web, IoT, credit card sources → central model

PROFESSIONAL DEFINITION

A hub-and-spoke architecture where diverse non-traditional data sources feed a central model.

KEY DETAILS

- Each spoke: cost, latency, legal status
- Competitive edge from unique combinations
- Growing data vendor ecosystem
- Due diligence on each source essential

ILLUSTRATIVE FIGURE

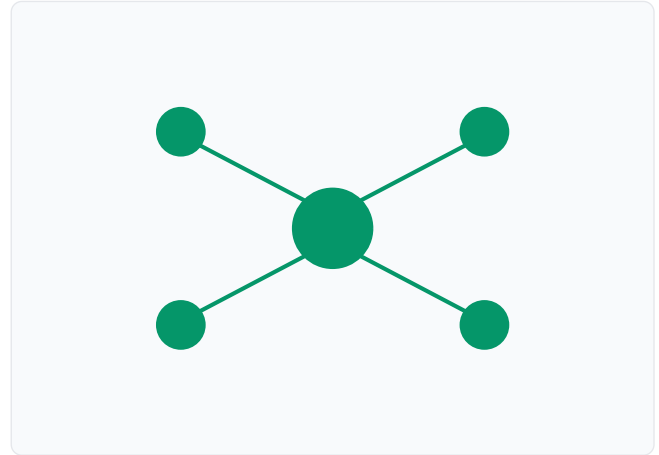


Figure 23.3 — *Satellite, social, web, IoT, credit card sources → central model*

SLIBRARY 23

Financial Analysis & Modelling

SOURCE (HARVARD REFERENCING STYLE)

Kolanovic, M. and Krishnamachari, R.T. (2017) Big Data and AI Strategies. J.P. Morgan.

NLP for Earnings Calls

SLIBRARY 23 · SHEET 4 / 5

Audio → transcript → entities → sentiment → signal

PROFESSIONAL DEFINITION

An automated pipeline: audio → transcript → entities → sentiment → predictive signal.

KEY DETAILS

- Management tone vs. analyst tone
- Hedging language signals caution
- Automated across thousands of calls
- Increasingly standard practice

ILLUSTRATIVE FIGURE

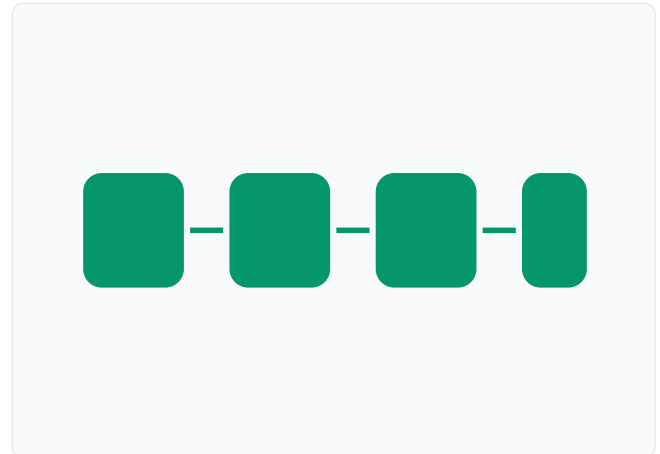


Figure 23.4 — *Audio → transcript → entities → sentiment → signal*

SLIBRARY 23

Financial Analysis & Modelling

SOURCE (HARVARD REFERENCING STYLE)

Loughran, T. and McDonald, B. (2016) 'Textual analysis in accounting and finance', *Journal of Accounting Research*, 54(4), pp. 1187–1230.

AI-Driven Risk Model Layers

SLIBRARY 23 · SHEET 5 / 5

Market → credit → operational risk with AI at each level

PROFESSIONAL DEFINITION

Stacked AI application to market risk, credit risk, and operational risk for integrated view.

KEY DETAILS

- Three stacked risk domains
- AI stress scenarios generated by ML
- Links to Slibrary 14 risk management
- Integrated risk view for leadership

ILLUSTRATIVE FIGURE

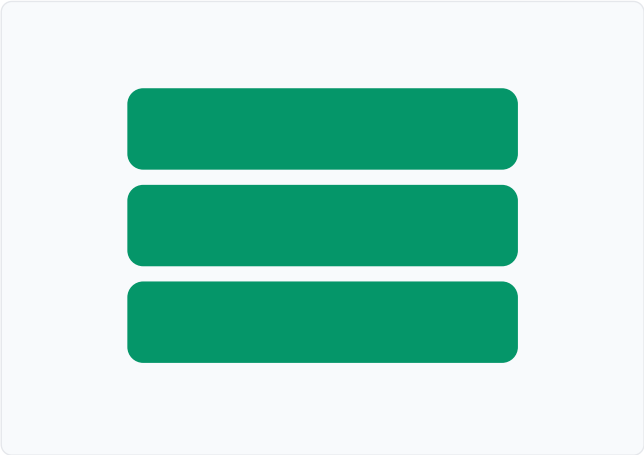


Figure 23.5 — Market → credit → operational risk with AI at each level

SLIBRARY 23

Financial Analysis & Modelling

SOURCE (HARVARD REFERENCING STYLE)

Hull, J.C. (2018) Risk Management and Financial Institutions. 5th edn. Hoboken: Wiley.

Planning vs. Implementation Gap

SLIBRARY 24 · SHEET 1 / 5

Ambition bar vs. delivery bar with barrier zone

PROFESSIONAL DEFINITION

A diagnostic showing the disparity between AI ambition and actual delivery, with barriers labelled.

KEY DETAILS

- Two bars: ambition (tall) vs. delivery (short)
- Gap labelled with common barriers
- Data, talent, governance, culture
- Core Unit 6 diagnostic thumbnail

ILLUSTRATIVE FIGURE

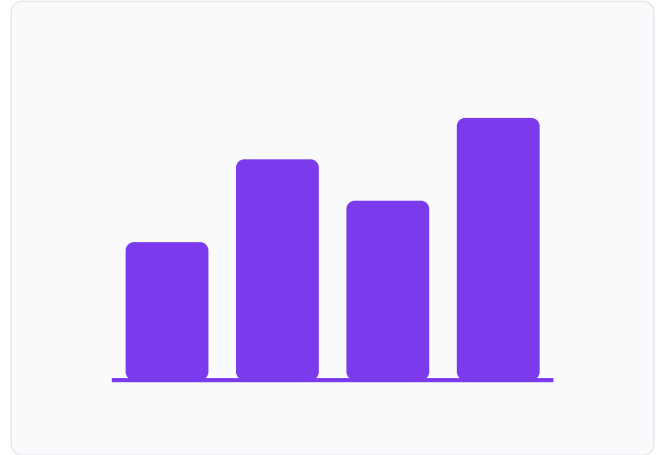


Figure 24.1 — Ambition bar vs. delivery bar with barrier zone

SLIBRARY 24

Plan-to-Implementation

SOURCE (HARVARD REFERENCING STYLE)

Ransbotham, S. et al. (2019) 'Winning with AI', MIT Sloan Management Review and BCG.

AI Readiness Assessment Radar

SLIBRARY 24 · SHEET 2 / 5

Spider chart: data, talent, tech, governance, culture axes

PROFESSIONAL DEFINITION

A spider chart evaluating preparedness across data, talent, technology, governance, and culture.

KEY DETAILS

- Current state vs. target overlay
- Identifies weakest spoke
- Actionable gap analysis at a glance
- Customisable to any organisation

ILLUSTRATIVE FIGURE

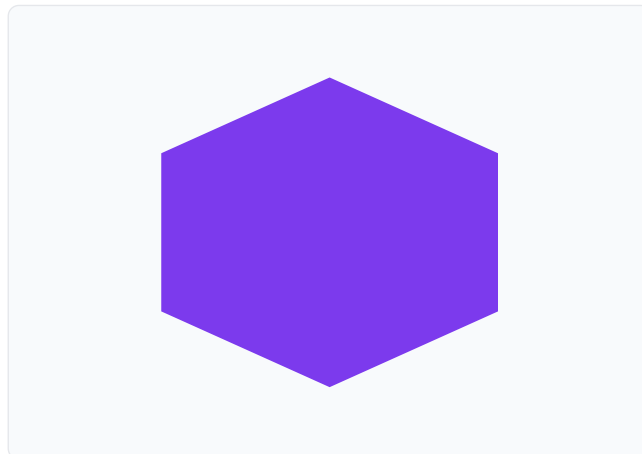


Figure 24.2 — *Spider chart: data, talent, tech, governance, culture axes*

SLIBRARY 24

Plan-to-Implementation

SOURCE (HARVARD REFERENCING STYLE)

Fountainne, T., McCarthy, B. and Saleh, T. (2019) 'Building the AI-powered organization', Harvard Business Review, 97(4), pp. 62–73.

Change Management Flywheel

SLIBRARY 24 · SHEET 3 / 5

Vision → quick wins → momentum → embedding cycle

PROFESSIONAL DEFINITION

A circular momentum model: vision → quick wins → momentum → embedding, each cycle reinforcing the next.

KEY DETAILS

- Each success feeds the next
- Demo value early to build trust
- Kotter-inspired, adapted for AI
- Continuous, not one-off transformation

ILLUSTRATIVE FIGURE



Figure 24.3 — *Vision → quick wins → momentum → embedding cycle*

SLIBRARY 24

Plan-to-Implementation

SOURCE (HARVARD REFERENCING STYLE)

Kotter, J.P. (2012) Leading Change. 2nd edn. Boston: Harvard Business Review Press. Adapted for AI.

Pilot-to-Scale Staircase

SLIBRARY 24 · SHEET 4 / 5

POC → pilot → production → enterprise with gates

PROFESSIONAL DEFINITION

A four-step progression—POC, pilot, production, enterprise—with governance gates between each.

KEY DETAILS

- Most AI projects stall at pilot
- Governance gates between steps
- Investment and risk increase per step
- Clear criteria for progression

ILLUSTRATIVE FIGURE

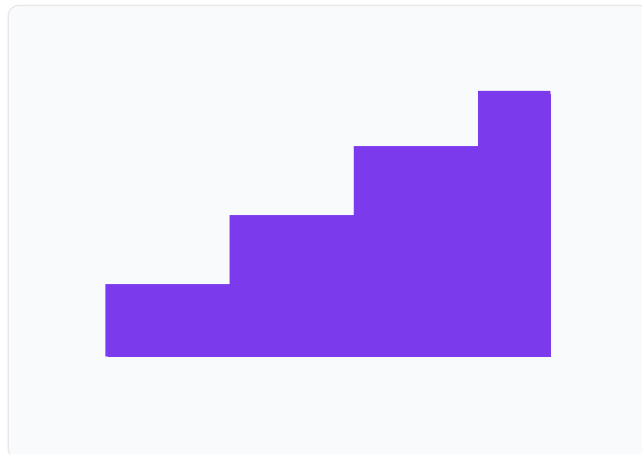


Figure 24.4 — *POC → pilot → production → enterprise with gates*

SLIBRARY 24

Plan-to-Implementation

SOURCE (HARVARD REFERENCING STYLE)

VentureBeat (2019) 'Why do 87% of data science projects never make it into production?', VentureBeat, 19 July.

Investment AI Maturity Model

SLIBRARY 24 · SHEET 5 / 5

Manual → assisted → augmented → autonomous

PROFESSIONAL DEFINITION

A four-level capability ladder: manual, assisted, augmented, autonomous.

KEY DETAILS

- Four capability levels
- Maps to org readiness radar
- Links to Slibrary 1 AI Maturity
- End-state vision for leadership

ILLUSTRATIVE FIGURE

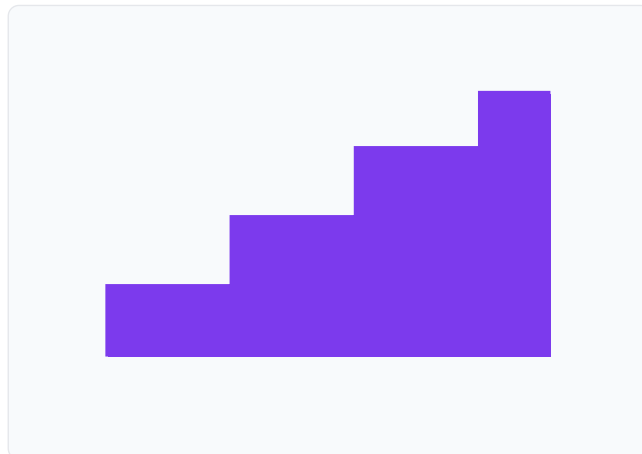


Figure 24.5 — *Manual → assisted → augmented → autonomous*

SLIBRARY 24

Plan-to-Implementation

SOURCE (HARVARD REFERENCING STYLE)

Muldowney, P. (2023) 'The impact of technology and AI on investment management', *Plans & Trusts*, 41(6), pp. 8–13.